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TRACKER MODULE

Abstract

A tracker module is provided that includes a module laminate, and an integrated circuit disposed at the module laminate. The integrated circuit includes at least one switch included in a pre-regulator circuit; at least one switch included in a switched-capacitor circuit; at least one switch included in an output switching circuit; and a digital control circuit. In plan view of the module laminate, at least one of the at least one switch included in the output switching circuit; or the digital control circuit is disposed between the at least one switch included in the pre-regulator circuit and the at least one switch included in the switched-capacitor circuit.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/JP2023/042240, filed Nov. 24, 2023, which claims priority to U.S. Provisional Patent Application No. 63/431,060, filed Dec. 8, 2022, the entire contents of each of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The exemplary aspects of the present disclosure relate to a tracker module.

BACKGROUND

[0003] In general, U.S. Pat. No. 9,755,672 discloses a supply modulator (e.g., a tracker circuit) configured to, based on an envelope signal, supply a power supply voltage to a power amplifier circuit. The supply modulator includes a magnetic converter circuit (magnetic regulation stage: pre-regulator circuit) configured to convert a voltage, a switched-capacitor circuit (switched-capacitor voltage balancer stage) configured to generate, from the voltage, a plurality of voltages having different voltage levels, and an output switching circuit (output switching stage) configured to select and output at least one of the plurality of voltages. The magnetic converter circuit includes switches and a power inductor. The switched-capacitor circuit includes switches and capacitors. Moreover, the output switching circuit includes switches.

[0004] With the configuration of the tracker circuit disclosed in U.S. Pat. No. 9,755,672, the switched-capacitor circuit generates a plurality of discrete voltages based on a voltage output from the pre-regulator circuit. As a result, the configuration may lead to efficiency degradation of the tracker circuit due to heat generation.

SUMMARY OF THE INVENTION

[0005] In view of the foregoing, the exemplary aspects of the present disclosure provide a tracker module that reduces efficiency degradation caused by heat generation.

[0006] According to an exemplary aspect, a tracker module is provided that includes a module laminate, and a single integrated circuit disposed at the module laminate. The integrated circuit includes at least one switch included in a converter circuit that is configured to convert an input voltage into a first voltage; at least one switch included in a switched-capacitor circuit that is configured to generate a plurality of discrete second voltages based on the first voltage; at least one switch included in an output switching circuit that is configured to selectively output at least one of the plurality of discrete second voltages generated by the switched-capacitor circuit; and a digital control circuit configured to, based on a digital control signal corresponding to an envelope signal, cause the output switching circuit to selectively output at least one of the plurality of discrete second voltages. In plan view of the module laminate, at least one of: (i) the at least one switch included in the output switching circuit; or (ii) the digital control circuit is disposed between the at least one switch included in the converter circuit and the at least one switch included in the switched-capacitor circuit.

[0007] In another exemplary aspect, a tracker module is provided that includes a module laminate, a first integrated circuit, a second integrated circuit, a third integrated circuit, and a digital control circuit. The first integrated circuit is disposed at the module laminate, and includes at least one switch included in a converter circuit, which is configured to convert an input voltage into a first voltage. The second integrated circuit is disposed at the module laminate, and includes at least one switch included in a switched-capacitor circuit, which is configured to generate a plurality of discrete second voltages based on the first voltage. The third integrated circuit is disposed at the

module laminate, and includes at least one switch included in an output switching circuit, which is configured to selectively output at least one of the plurality of discrete second voltages generated by the switched-capacitor circuit. The digital control circuit is configured to, based on a digital control signal corresponding to an envelope signal, cause the output switching circuit to selectively output at least one of the plurality of discrete second voltages. In a plan view of the module laminate, at least one of: (i) the at least one switch included in the output switching circuit; or (ii) the digital control circuit is disposed between the at least one switch included in the converter circuit and the at least one switch included in the switched-capacitor circuit.

[0008] In another exemplary aspect, a tracker module is provided that includes a module laminate, and a single integrated circuit disposed at the module laminate. The integrated circuit includes a first switch, a second switch, a third switch, a fourth switch, a fifth switch, a sixth switch, a seventh switch, and an eighth switch that are included in a switched-capacitor circuit; a ninth switch and a tenth switch that are included in an output switching circuit connected to the switched-capacitor circuit; an eleventh switch and a twelfth switch that are included in a converter circuit connected to the switched-capacitor circuit; and a digital control circuit configured to control the first to twelfth switches. The switched-capacitor circuit includes a first capacitor having a first electrode and a second electrode; a second capacitor having a third electrode and a fourth electrode; and the first to eighth switches. A first terminal of the first switch and a first terminal of the third switch are connected to the first electrode. A first terminal of the second switch and a first terminal of the fourth switch are connected to the second electrode. A first terminal of the fifth switch and a first terminal of the seventh switch are connected to the third electrode. A first terminal of the sixth switch and a first terminal of the eighth switch are connected to the fourth electrode. A second terminal of the first switch, a second terminal of the second switch, a second terminal of the fifth switch, and a second terminal of the sixth switch are connected to each other. A second terminal of the third switch is connected to a second terminal of the seventh switch. A second terminal of the fourth switch is connected to a second terminal of the eighth switch. The output switching circuit includes: a first output terminal; the ninth switch connected between the first output terminal, and the second terminal of the first switch, the second terminal of the second switch, the second terminal of the fifth switch, and the second terminal of the sixth switch; and the tenth switch connected between the first output terminal, and the second terminal of the third switch and the second terminal of the seventh switch. The converter circuit includes: an input terminal; the eleventh switch connected between the input terminal and a first end of a power inductor; and the twelfth switch connected between the first end of the power inductor and ground. A second end of the power inductor is connected to the second terminal of the first switch, the second terminal of the second switch, the second terminal of the fifth switch, and the second terminal of the sixth switch. In a plan view of the module laminate, at least one of: (i) the ninth switch and the tenth switch, and (ii) the digital control circuit is disposed between: the eleventh switch and the twelfth switch; and the first switch, the second switch, the third switch, the fourth switch, the fifth switch, the sixth switch, the seventh switch, and the eighth switch.

[0009] The exemplary aspects of the present disclosure provide a tracker module that reduces efficiency degradation caused by heat generation.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. 1A is a graph illustrating an example of the progression of power supply voltage in an average power tracking (APT) mode.

[0011] FIG. 1B is a graph illustrating an example of the progression of power supply voltage in an analog ET mode.

[0012] FIG. 1C is a graph illustrating an example of the progression of power supply voltage in a digital ET mode.

[0013] FIG. 2 is a circuit diagram of a tracker circuit and a communication device according to exemplary embodiments.

[0014] FIG. 3 is a circuit diagram of a pre-regulator circuit, a switched-capacitor circuit, an output switching circuit, and a filter circuit according to the exemplary embodiments.

[0015] FIG. 4 is a circuit diagram of a digital control circuit according to the exemplary embodiments.

[0016] FIG. 5A is a plan view of a tracker module according to an example.

[0017] FIG. 5B is a plan view of the tracker module according to an example.

[0018] FIG. 5C is a cross-sectional view of the tracker module according to an example.

[0019] FIG. 5D is a plan view of a portion of an integrated circuit according to an example.

[0020] FIG. 6A is a plan view of a tracker module according to Modification 1.

[0021] FIG. 6B is a cross-sectional view of the tracker module according to 1.

[0022] FIG. 7 is a plan view of a tracker module according to Modification 2.

[0023] FIG. 8 is a plan view of a tracker module according to Modification 3.

[0024] FIG. 9A is a plan view of a tracker module according to Modification 4.

[0025] FIG. 9B is a cross-sectional view of the tracker module according to Modification 4.

[0026] FIG. 10 is a plan view of a tracker module according to Modification 5.

[0027] FIG. 11A is a plan view of a tracker module according to Modification 6.

[0028] FIG. 11B is a plan view of the tracker module according to Modification 6.

[0029] FIG. 11C is a cross-sectional view of the tracker module according to Modification 6.

DETAILED DESCRIPTION OF EMBODIMENTS

[0030] Exemplary embodiments of the present disclosure will now be described in detail below with reference to the drawings. The embodiments described below each represent generic or specific examples. Features presented in the following embodiments, such as numerical values, shapes, materials, constituent elements, and the positioning and connection of constituent elements, are illustrative only and not intended to be limiting of the exemplary aspects of the present disclosure.

[0031] The drawings are schematic in nature with emphases, omissions, or proportion adjustments made as necessary to illustrate the exemplary aspects of the present disclosure, and do not necessarily represent exact details. Accordingly, the illustrated shapes, positional relationships, and proportions may differ from the actuality. Throughout the drawings, identical reference signs are used to designate substantially identical features, and repetitive description will be sometimes omitted or simplified.

[0032] In the figures described below, the x-axis and the y-axis are orthogonal to each other in a plane parallel to a major face of a module laminate. Specifically, when the module laminate has a rectangular shape in plan view, the x-axis is parallel to a first side of the module laminate, and the y-axis is parallel to a second side orthogonal to the first side of the module laminate. A z-axis is an axis perpendicular to the major face of the module laminate. The z-axis has a positive direction defined as an upward direction, and a negative direction defined as a downward direction.

[0033] As used in the following description of circuit configurations according to the present disclosure, expressions such as “connected” can indicate not only that circuit elements are directly connected by a connection terminal and/or a wiring conductor, but also that circuit elements are electrically connected with another circuit element interposed therebetween. Expressions such as “connected between A and B” can refer to being positioned between A and B and connected to both A and B.

[0034] As used herein with regard to the positioning of components according to the exemplary aspects of the present disclosure, expressions such as “a component is disposed at a substrate” include that the component is disposed on a major face of the substrate, and that the component is

disposed in the substrate. Expressions such as “a component is disposed on a major face of a substrate” include not only that the component is disposed in contact with the major face of the substrate, but also that the component is disposed above the major face without making contact with the major face (e.g., the component is stacked on another component disposed in contact with the major face). The expressions such as “a component is disposed on a major face of a substrate” may also include that the component is disposed at a recess defined in the major face. Expressions such as “a component is disposed in a substrate” include, in addition to the meaning that the component is encapsulated in the module laminate, the following meanings: the entirety of the component is disposed between opposite major faces of the substrate but part of the component is not covered by the substrate; and only part of the component is disposed in the substrate.

[0035] As used herein with regard to the positioning of components according to the exemplary aspects of the present disclosure, expressions such as “plan view of a module laminate” can indicate that an object or component is orthogonally projected onto an xy-plane and seen from the positive side of the z-axis. Expressions such as “A overlaps B in plan view” can indicate that at least part of the region of A orthogonally projected onto the xy-plane overlaps at least part of the region of B orthogonally projected onto the xy-plane. Expressions such as “A is disposed between B and C” can indicate that at least one of a plurality of line segments each connecting a given point in B and a given point in C passes through A.

[0036] As used herein with regard to the positioning of components according to the exemplary aspects of the present disclosure, expressions such as “A is disposed adjacent to B” indicate that A and B are disposed in proximity to each other. Specifically, such expressions can indicate that no other circuit component exists in a space where A faces B. In other words, the expressions such as “A is disposed adjacent to B” can indicate that none of a plurality of line segments each extending from a given point on a surface of A facing B to B in the direction normal to the surface passes through a circuit component other than A and B. In this regard, a circuit component can refer to a component including an active element and/or a passive element. That is, examples of such circuit components include active components such as transistors or diodes, and passive components such as inductors, transformers, capacitors, or resistors, but do not include electromechanical components such as terminals, connectors, or wiring.

[0037] As used in the exemplary aspects of the present disclosure, expressions such as “terminal” can refer to a point where a conductor within an element terminates. When the impedance of a conductor located between elements is sufficiently low, a terminal is interpreted not only as a single point, but also as any given point on the conductor located between the elements or as the entire conductor.

[0038] Further, “parallel”, “perpendicular”, or other such expressions indicative of the relationship between elements, and “rectangular” or other such expressions indicative of a shape of an element, as well as numerical ranges are not intended to represent only their strict meanings but are meant to also include their substantial equivalents, for example, equivalents with deviations or differences of about several percent.

[0039] First, as a technique for efficiently amplifying a radio frequency signal, a tracking mode is described below in which a power amplifier receives a power supply voltage that is dynamically adjusted with the passage of time based on the radio frequency signal. A tracking mode refers to a mode that dynamically adjusts the power supply voltage to be applied to a power amplifier. Several types of tracking modes exist, of which an average power tracking (APT) mode and an envelope tracking (ET) mode (including an analog ET mode and a digital ET mode) will now be described with reference to FIGS. 1A to 1C. In FIGS. 1A to 1C, the horizontal axis represents time, and the vertical axis represents voltage. A thick solid line represents power supply voltage, and a thin solid line (waveform) represents modulated signal.

[0040] FIG. 1A is a graph illustrating an example of the progression of power supply voltage in the APT mode. In the APT mode, the power supply voltage is varied across a plurality of discrete

voltage levels in units of one frame based on average power. As a result, a power supply voltage signal forms a rectangular wave.

[0041] According to an exemplary aspect, a frame can refer to a unit forming a radio frequency signal (e.g., a modulated signal). For example, in 5th Generation New Radio (5G NR) and Long Term Evolution (LTE), a frame includes ten subframes. Each subframe includes a plurality of slots. Each slot includes a plurality of symbols. A subframe has a length of 1 ms, and a frame has a length of 10 ms according to an exemplary aspect.

[0042] According to an exemplary aspect, a mode in which the voltage level is varied in units of one frame or in larger units based on average power is referred to as an APT mode, which is distinguished from a mode in which the voltage level is varied in units smaller than one frame (e.g., in units of subframes, slots, or symbols). For example, a mode in which the voltage level is varied in units of symbols is referred to as a symbol power tracking (SPT) mode, which is distinguished from the APT mode.

[0043] FIG. 1B is a graph illustrating an example of the progression of power supply voltage in the analog ET mode. In the analog ET mode, the power supply voltage is varied continuously based on an envelope signal to thereby track the envelope of a modulated signal.

[0044] An envelope signal is a signal representing the envelope of a modulated signal. An envelope value is expressed as, for example, the square root of $(I_{\text{sup}}^2 + Q_{\text{sup}}^2)$. In this case, (I, Q) represents a constellation point. A constellation point is a point representing, on a constellation diagram, a signal modulated by digital modulation. (I, Q) is determined by, for example, a baseband integrated circuit (BBIC) based on, for example, transmission information.

[0045] FIG. 1C is a graph illustrating an example of the progression of power supply voltage in the digital ET mode. In the digital ET mode, the power supply voltage is varied across a plurality of discrete voltage levels within one frame based on an envelope signal to thereby track the envelope of a modulated signal. As a result, the power supply voltage signal forms a rectangular wave.

Exemplary Embodiments

[0046] A communication device **6** according to the embodiments corresponds to user equipment (UE) in a cellular network. Typical examples of the communication device **6** include a mobile phone, a smartphone, a tablet computer, and a wearable device. The communication device **6** may be an Internet of Things (IoT) sensor device, a medical/healthcare device, a vehicle, an unmanned aerial vehicle (UAV) (a so-called drone), or an automated guided vehicle (AGV). The communication device **6** can be configured as a base station (BS) in a cellular network according to an exemplary aspect.

[0047] The circuit configuration of each of the communication device **6** and a tracker circuit **1** according to the embodiments will now be described with reference to FIG. 2. FIG. 2 is a circuit diagram of the tracker circuit **1** and the communication device **6** according to the embodiments.

[0048] FIG. 2 illustrates an exemplary circuit configuration. The communication device **6** and the tracker circuit **1** may be implemented by using any one of a wide variety of circuit implementations and circuit technologies. Therefore, the description of the communication device **6** and the tracker circuit **1** provided below is not to be construed restrictively.

1 Circuit Configuration of Communication Device 6

[0049] First, the communication device **6** according to the embodiments will be described with reference to FIG. 2. The communication device **6** includes the tracker circuit **1**, a power amplifier **2**, a filter **3**, a radio frequency integrated circuit (RFIC) **5**, and an antenna **4**.

[0050] According to an exemplary aspect, the tracker circuit **1** is configured to, based on a tracking mode, supply a plurality of discrete voltages V_A to the power amplifier **2**. The tracking mode used may be, but is not limited to, the digital ET mode or the SPT mode. As illustrated in FIG. 2, the tracker circuit **1** includes a pre-regulator circuit **10**, a switched-capacitor circuit **20**, an output switching circuit **30**, a filter circuit **40**, a direct current (DC) power source **50**, and a digital control circuit **60**.

[0051] The pre-regulator circuit **10** is an example of a converter circuit, and includes a power inductor and a switch. A power inductor is an inductor used to step up and/or step down a DC voltage. The power inductor is connected in series with a DC path. The power inductor may be connected (disposed in parallel) between the DC path and ground. The pre-regulator circuit **10** is configured to convert an input voltage into a first voltage by using the power inductor. According to an exemplary aspect, the pre-regulator circuit **10** may also be referred to as a magnetic regulator or a DC-DC converter.

[0052] The switched-capacitor circuit **20** includes a plurality of capacitors, and a plurality of switches. The switched-capacitor circuit **20** is configured to receive the first voltage from the pre-regulator circuit **10** and, from the first voltage, generate, as a plurality of discrete voltages, a plurality of second voltages each having a corresponding one of a plurality of discrete voltage levels. According to an exemplary aspect, the switched-capacitor circuit **20** may also be referred to as a switched-capacitor voltage balancer.

[0053] The output switching circuit **30** is configured to selectively output, to the power amplifier **2**, at least one of the plurality of second voltages generated by the switched-capacitor circuit **20**. The output switching circuit **30** is controlled based on a digital control signal.

[0054] The filter circuit **40** is configured to attenuate noise from the plurality of discrete voltages to be supplied to the power amplifier **2**. According to an exemplary aspect, the filter circuit **40** may also be referred to as a pulse shaping filter or a transition shaping filter.

[0055] According to an exemplary aspect, the DC power source **50** is configured to supply a DC voltage to the pre-regulator circuit **10**. A suitable, non-limiting example of the DC power source **50** is a rechargeable battery.

[0056] The digital control circuit **60** is configured to control the following components based on a digital control signal provided from the RFIC **5**: the pre-regulator circuit **10**, the switched-capacitor circuit **20**, the output switching circuit **30**, and the filter circuit **40**. More specifically, the digital control circuit **60** is configured to, based on a digital control signal corresponding to an envelope signal, control the output switching circuit **30** to selectively output at least one of the plurality of discrete second voltages.

[0057] The tracker circuit **1** may be configured not to include at least one of the filter circuit **40** or the DC power source **50**. In one example, the tracker circuit **1** may be configured not to include the DC power source **50**. In another example, any combination of the pre-regulator circuit **10**, the switched-capacitor circuit **20**, the output switching circuit **30**, and the filter circuit **40** may be integrated into a single circuit. In another example, the tracker circuit **1** may include a plurality of voltage supply circuits, instead of the pre-regulator circuit **10** and the switched-capacitor circuit **20**. In this case, the output switching circuit **30** may be configured to select at least one of the plurality of voltage supply circuits.

[0058] The power amplifier **2** is connected between the RFIC **5** and the filter **3**. Further, the power amplifier **2** is connected to the tracker circuit **1**. The power amplifier **2** is configured to amplify, by using the plurality of discrete voltages V_A received from the tracker circuit **1**, a radio frequency signal RFA of Band A received from the RFIC **5**.

[0059] The filter **3** is connected between the power amplifier **2** and the antenna **4**. The filter **3** is a band pass filter with a passband that includes Band A.

[0060] Band A is a frequency band for communication systems built by using radio access technology (RAT). Band A is predefined by standardizing bodies or other entities (e.g., 3rd Generation Partnership Project (3GPP)[®] and Institute of Electrical and Electronics Engineers (IEEE)). Examples of such communication systems include 5th Generation New Radio (5G NR) systems, Long Term Evolution (LTE) systems, and Wireless Local Area Network (WLAN) systems.

[0061] The RFIC **5** is an example of a signal processing circuit that processes a radio frequency signal. Specifically, the RFIC **5** performs signal processing such as up-conversion on an input

transmission signal to thereby generate a radio frequency transmission signal, and supplies the radio frequency transmission signal to the power amplifier 2. The RFIC 5 includes a control unit that controls the tracker circuit 1. The function of the RFIC 5 as a control unit may in part or in whole be implemented outside the RFIC 5 in an exemplary aspect.

[0062] The antenna 4 outputs a transmission signal of Band A received from the power amplifier 2 via the filter 3. The antenna 4 need not necessarily be included in the communication device 6.

[0063] The circuit configuration of the communication device 6 in FIG. 2 is illustrative and not intended to be limiting. For example, the communication device 6 may include a baseband signal processing circuit that performs signal processing by using an intermediate frequency band lower in frequency than the radio frequency signal RFA.

2 Circuit Configuration of Tracker Circuit 1

[0064] Now, with reference to FIGS. 3 and 4, the circuit configuration of each of the following components included in the tracker circuit 1 will be described: the pre-regulator circuit 10, the switched-capacitor circuit 20, the output switching circuit 30, the filter circuit 40, and the digital control circuit 60. FIG. 3 is a circuit diagram of the following components according to the embodiments: the pre-regulator circuit 10, the switched-capacitor circuit 20, the output switching circuit 30, and the filter circuit 40. FIG. 4 is a circuit diagram of the digital control circuit 60 according to the embodiments.

[0065] FIGS. 3 and 4 illustrate exemplary circuit configurations. The pre-regulator circuit 10, the switched-capacitor circuit 20, the output switching circuit 30, the filter circuit 40, and the digital control circuit 60 may be implemented by using any one of a wide variety of circuit implementations and circuit technologies. Therefore, the description of each circuit provided below is not to be construed restrictively.

2.1 Circuit Configuration of Switched-Capacitor Circuit 20

[0066] First, the circuit configuration of the switched-capacitor circuit 20 is described with reference to FIG. 3. As illustrated in FIG. 3, the switched-capacitor circuit 20 includes capacitors C11 to C16, capacitors C10, C20, C30 and C40, and switches S11 to S14, S21 to S24, S31 to S34, and S41 to S44. Energy and charge are input, at nodes N1 to N4, from the pre-regulator circuit 10 to the switched-capacitor circuit 20, and drawn, at the nodes N1 to N4, from the switched-capacitor circuit 20 to the output switching circuit 30.

[0067] Each of the capacitors C11 to C16 can be configured to function as a flying capacitor (sometimes also referred to as a transfer capacitor) in an exemplary aspect. That is, each of the capacitors C11 to C16 is used to step up or step down the first voltage supplied from the pre-regulator circuit 10. More specifically, the capacitors C11 to C16 transfer charge between the capacitors C11 to C16 and the nodes N1 to N4 so that voltages V1 to V4 (relative to the ground potential) that satisfy $V1:V2:V3:V4=1:2:3:4$ are maintained at the four nodes N1 to N4. The voltages V1 to V4 correspond to the plurality of second voltages each having a corresponding one of a plurality of discrete voltage levels.

[0068] The capacitor C11 is an example of a first capacitor, and has two electrodes. One of the two electrodes of the capacitor C11 is an example of a first electrode, and is connected to one terminal of the switch S11 and one terminal of the switch S12. The other of the two electrodes of the capacitor C11 is an example of a second electrode, and is connected to one terminal of the switch S21 and one terminal of the switch S22. For purposes of the exemplary aspects of the present disclosure, the term “one terminal” can be considered or referred to as a first terminal.

[0069] The capacitor C12 has two electrodes. One of the two electrodes of the capacitor C12 is connected to the one terminal of the switch S21 and the one terminal of the switch S22. The other of the two electrodes of the capacitor C12 is connected to one terminal of the switch S31 and one terminal of the switch S32.

[0070] The capacitor C13 has two electrodes. One of the two electrodes of the capacitor C13 is connected to the one terminal of the switch S31 and the one terminal of the switch S32. The other

of the two electrodes of the capacitor C13 is connected to one terminal of the switch S41 and one terminal of the switch S42.

[0071] The capacitor C14 is an example of a second capacitor, and has two electrodes. One of the two electrodes of the capacitor C14 is an example of a third electrode, and is connected to one terminal of the switch S13 and one terminal of the switch S14. The other of the two electrodes of the capacitor C14 is an example of a fourth electrode, and is connected to one terminal of the switch S23 and one terminal of the switch S24.

[0072] The capacitor C15 has two electrodes. One of the two electrodes of the capacitor C15 is connected to the one terminal of the switch S23 and the one terminal of the switch S24. The other of the two electrodes of the capacitor C15 is connected to one terminal of the switch S33 and one terminal of the switch S34.

[0073] The capacitor C16 has two electrodes. One of the two electrodes of the capacitor C16 is connected to the one terminal of the switch S33 and the one terminal of the switch S34. The other of the two electrodes of the capacitor C16 is connected to one terminal of the switch S43 and one terminal of the switch S44.

[0074] A set of the capacitors C11 and C14, a set of the capacitors C12 and C15, and a set of the capacitors C13 and C16 are each configured to be charged and discharged in a complementary manner as a first phase and a second phase are repeated.

[0075] Specifically, in the first phase, the switches S12, S13, S22, S23, S32, S33, S42, and S43 are turned ON. As a result, for example, one of the two electrodes of the capacitor C12 is connected to the node N3, the other of the two electrodes of the capacitor C12 and the one of the two electrodes of the capacitor C15 are connected to the node N2, and the other of the two electrodes of the capacitor C15 is connected to the node N1.

[0076] In the second phase, the switches S11, S14, S21, S24, S31, S34, S41, and S44 are turned ON. As a result, for example, the one of the two electrodes of the capacitor C15 is connected to the node N3, the other of the two electrodes of the capacitor C15 and the one of the two electrodes of the capacitor C12 are connected to the node N2, and the other of the two electrodes of the capacitor C12 is connected to the node N1.

[0077] The first phase and the second phase are repeated as described above. Consequently, for example, when one of the capacitors C12 and C15 is being charged from the node N2, the other of the capacitors C12 and C15 can be discharged to the capacitor C30. That is, the capacitors C12 and C15 are configured to charge and discharge in a complementary manner.

[0078] Likewise, as with the set of the capacitors C12 and C15, the set of the capacitors C11 and C14 and the set of the capacitors C13 and C16 are each configured to charge and discharge in a complementary manner as the first phase and the second phase are repeated.

[0079] Each of the capacitors C10, C20, C30, and C40 can be configured to function as a smoothing capacitor in an exemplary aspect. That is, the capacitors C10, C20, C30, and C40 are respectively used to hold and smooth the voltages V1 to V4 at the nodes N1 to N4.

[0080] The capacitor C10 is connected between the node N1 and ground. Specifically, one of two electrodes of the capacitor C10 is connected to the node N1. The other of the two electrodes of the capacitor C10 is connected to ground.

[0081] The capacitor C20 is connected between the nodes N2 and N1. Specifically, one of two electrodes of the capacitor C20 is connected to the node N2. The other of the two electrodes of the capacitor C20 is connected to the node N1.

[0082] The capacitor C30 is connected between the nodes N3 and N2. Specifically, one of two electrodes of the capacitor C30 is connected to the node N3. The other of the two electrodes of the capacitor C30 is connected to the node N2.

[0083] The capacitor C40 is connected between the nodes N4 and N3. Specifically, one of two electrodes of the capacitor C40 is connected to the node N4. The other of the two electrodes of the capacitor C40 is connected to the node N3.

[0084] The switch **S11** is an example of a first switch, and is connected between the one of the two electrodes of the capacitor **C11** and the node **N3**. Specifically, the one terminal of the switch **S11** is connected to the one of the two electrodes of the capacitor **C11**. The other terminal of the switch **S11** is connected to the node **N3**. For purposes of the exemplary aspects of the present disclosure, the term “other terminal” can be considered or referred to as a second terminal.

[0085] The switch **S12** is an example of a third switch, and is connected between the one of the two electrodes of the capacitor **C11** and the node **N4**. Specifically, the one terminal of the switch **S12** is connected to the one of the two electrodes of the capacitor **C11**. The other terminal of the switch **S12** is connected to the node **N4**.

[0086] The switch **S21** is an example of a fourth switch, and is connected between the one of the two electrodes of the capacitor **C12** and the node **N2**. Specifically, the one terminal of the switch **S21** is connected to the one of the two electrodes of the capacitor **C12** and the other of the two electrodes of the capacitor **C11**. The other terminal of the switch **S21** is connected to the node **N2**.

[0087] The switch **S22** is an example of a second switch, and is connected between the one of the two electrodes of the capacitor **C12** and the node **N3**. Specifically, the one terminal of the switch **S22** is connected to the one of the two electrodes of the capacitor **C12** and the other of the two electrodes of the capacitor **C11**. The other terminal of the switch **S22** is connected to the node **N3**.

[0088] The switch **S31** is connected between the other of the two electrodes of the capacitor **C12** and the node **N1**. Specifically, the one terminal of the switch **S31** is connected to the other of the two electrodes of the capacitor **C12** and the one of the two electrodes of the capacitor **C13**. The other terminal of the switch **S31** is connected to the node **N1**.

[0089] The switch **S32** is connected between the other of the two electrodes of the capacitor **C12** and the node **N2**. Specifically, the one terminal of the switch **S32** is connected to the other of the two electrodes of the capacitor **C12** and the one of the two electrodes of the capacitor **C13**. The other terminal of the switch **S32** is connected to the node **N2**. That is, the other terminal of the switch **S32** is connected to the other terminal of the switch **S21**.

[0090] The switch **S41** is connected between the other of the two electrodes of the capacitor **C13** and ground. Specifically, the one terminal of the switch **S41** is connected to the other of the two electrodes of the capacitor **C13**. The other terminal of the switch **S41** is connected to ground.

[0091] The switch **S42** is connected between the other of the two electrodes of the capacitor **C13** and the node **N1**. Specifically, the one terminal of the switch **S42** is connected to the other of the two electrodes of the capacitor **C13**. The other terminal of the switch **S42** is connected to the node **N1**. That is, the other terminal of the switch **S42** is connected to the other terminal of the switch **S31**.

[0092] The switch **S13** is an example of a fifth switch, and is connected between the one of the two electrodes of the capacitor **C14** and the node **N3**. Specifically, the one terminal of the switch **S13** is connected to the one of the two electrodes of the capacitor **C14**. The other terminal of the switch **S13** is connected to the node **N3**. That is, the other terminal of the switch **S13** is connected to the other terminal of the switch **S11** and the other terminal of the switch **S22**.

[0093] The switch **S14** is an example of a seventh switch, and is connected between the one of the two electrodes of the capacitor **C14** and the node **N4**. Specifically, the one terminal of the switch **S14** is connected to the one of the two electrodes of the capacitor **C14**. The other terminal of the switch **S14** is connected to the node **N4**. That is, the other terminal of the switch **S14** is connected to the other terminal of the switch **S12**.

[0094] The switch **S23** is an example of an eighth switch, and is connected between the one of the two electrodes of the capacitor **C15** and the node **N2**. Specifically, the one terminal of the switch **S23** is connected to one of the two electrodes of the capacitor **C15** and the other of the two electrodes of the capacitor **C14**. The other terminal of the switch **S23** is connected to the node **N2**. That is, the other terminal of the switch **S23** is connected to the other terminal of the switch **S21** and the other terminal of the switch **S32**.

[0095] The switch **S24** is an example of a sixth switch, and is connected between the one of the two electrodes of the capacitor **C15** and the node **N3**. Specifically, the one terminal of the switch **S24** is connected to the one of the two electrodes of the capacitor **C15** and the other of the two electrodes of the capacitor **C14**. The other terminal of the switch **S24** is connected to the node **N3**. That is, the other terminal of the switch **S24** is connected to the other terminal of the switch **S11**, the other terminal of the switch **S22**, and the other terminal of the switch **S13**.

[0096] The switch **S33** is connected between the other of the two electrodes of the capacitor **C15** and the node **N1**. Specifically, the one terminal of the switch **S33** is connected to the other of the two electrodes of the capacitor **C15** and the one of the two electrodes of the capacitor **C16**. The other terminal of the switch **S33** is connected to the node **N1**. That is, the other terminal of the switch **S33** is connected to the other terminal of the switch **S31** and the other terminal of the switch **S42**.

[0097] The switch **S34** is connected between the other of the two electrodes of the capacitor **C15** and the node **N2**. Specifically, the one terminal of the switch **S34** is connected to the other of the two electrodes of the capacitor **C15** and the one of the two electrodes of the capacitor **C16**. The other terminal of the switch **S34** is connected to the node **N2**. That is, the other terminal of the switch **S34** is connected to the other terminal of the switch **S21**, the other terminal of the switch **S32**, and the other terminal of the switch **S23**.

[0098] The switch **S43** is connected between the other of the two electrodes of the capacitor **C16** and ground. Specifically, the one terminal of the switch **S43** is connected to the other of the two electrodes of the capacitor **C16**. The other terminal of the switch **S43** is connected to ground.

[0099] The switch **S44** is connected between the other of the two electrodes of the capacitor **C16** and the node **N1**. Specifically, the one terminal of the switch **S44** is connected to the other of the two electrodes of the capacitor **C16**. The other terminal of the switch **S44** is connected to the node **N1**. That is, the other terminal of the switch **S44** is connected to the other terminal of the switch **S31**, the other terminal of the switch **S42**, and the other terminal of the switch **S33**.

[0100] The switches **S11**, **S12**, **S13**, **S14**, **S21**, **S22**, **S23**, and **S24** correspond to at least one switch included in the switched-capacitor circuit **20**.

[0101] A first set of switches including the switches **S12**, **S13**, **S22**, **S23**, **S32**, **S33**, **S42**, and **S43**, and a second set of switches including the switches **S11**, **S14**, **S21**, **S24**, **S31**, **S34**, **S41**, and **S44** are switched between ON and OFF in a complementary manner based on a control signal **S2**.

Specifically, in the first phase, the switches in the first set are turned ON, and the switches in the second set are turned OFF. Conversely, in the second phase, the switches in the first set are turned OFF, and the switches in the second set are turned ON.

[0102] For example, in one of the first and second phases, charging of the capacitors **C10** to **C40** from the capacitors **C11** to **C13** is executed, and in the other of the first and second phases, charging of the capacitors **C10** to **C40** from the capacitors **C14** to **C16** is executed. That is, the capacitors **C10** to **C40** are constantly charged from the capacitors **C11** to **C13** or the capacitors **C14** to **C16**. This ensures that even when current flows rapidly from the nodes **N1** to **N4** to the output switching circuit **30**, the nodes **N1** to **N4** are rapidly replenished with charge. This configuration in turn reduces fluctuations in potential at the nodes **N1** to **N4**.

[0103] The operation mentioned above allows a substantially equal voltage to be maintained across each of the capacitors **C10**, **C20**, **C30**, and **C40** in the switched-capacitor circuit **20**. Specifically, the voltages **V1** to **V4** (relative to the ground potential) that satisfy $V1:V2:V3:V4 = 1:2:3:4$ are maintained at the four nodes labeled **V1** to **V4**. The voltage levels of the voltages **V1** to **V4** correspond to the plurality of discrete levels of voltage that can be supplied by the switched-capacitor circuit **20** to the output switching circuit **30**.

[0104] The voltage ratio $V1:V2:V3:V4$ is not limited to 1:2:3:4. For example, the voltage ratio $V1:V2:V3:V4$ may be 1:2:4:8.

[0105] The configuration of the switched-capacitor circuit **20** in FIG. 3 is illustrative and not

intended to be limiting. In FIG. 3, the switched-capacitor circuit **20** is configured to supply four discrete levels of voltage. This configuration, however, is not intended to be limiting. The switched-capacitor circuit **20** may be configured to supply any number of discrete levels of voltage greater than or equal to two. For example, in an exemplary aspect, when the switched-capacitor circuit **20** are configured to supply two discrete levels of voltage, it may suffice that the switched-capacitor circuit **20** includes at least the capacitors **C12** and **C15** and the switches **S21** to **S24** and **S31** to **S34**.

2.2 Circuit Configuration of Output Switching Circuit **30**

[0106] The circuit configuration of the output switching circuit **30** will now be described with reference to FIG. 3. The output switching circuit **30** is connected to the digital control circuit **60**. As illustrated in FIG. 3, the output switching circuit **30** includes input terminals **131** to **134**, switches **S51** to **S54**, and an output terminal **130**.

[0107] The output terminal **130** is connected to an input terminal **140** of the filter circuit **40**. The output terminal **130** is a terminal for supplying a power supply voltage selected from among the voltages **V1** to **V4** to the power amplifier **2** via the filter circuit **40**.

[0108] The input terminals **131** to **134** are respectively connected to the nodes **N4** to **N1** of the switched-capacitor circuit **20**. The input terminals **131** to **134** are terminals for respectively receiving the voltages **V4** to **V1** from the switched-capacitor circuit **20**.

[0109] The switch **S51** is an example of a tenth switch, and is connected between the input terminal **131** and the output terminal **130**. Specifically, the switch **S51** has a terminal connected to the input terminal **131**, and a terminal connected to the output terminal **130**. With the connection arrangement mentioned above, the switch **S51** is switched between ON and OFF based on a control signal **S3** to allow switching between connection and disconnection of the input terminal **131** and the output terminal **130** to and from each other.

[0110] The switch **S52** is an example of a ninth switch, and is connected between the input terminal **132** and the output terminal **130**. Specifically, the switch **S52** has a terminal connected to the input terminal **132**, and a terminal connected to the output terminal **130**. With the connection arrangement mentioned above, the switch **S52** is switched between ON and OFF based on the control signal **S3** to allow switching between connection and disconnection of the input terminal **132** and the output terminal **130** to and from each other.

[0111] The switch **S53** is connected between the input terminal **133** and the output terminal **130**. Specifically, the switch **S53** has a terminal connected to the input terminal **133**, and a terminal connected to the output terminal **130**. With the connection arrangement mentioned above, the switch **S53** is switched between ON and OFF based on the control signal **S3** to allow switching between connection and disconnection of the input terminal **133** and the output terminal **130** to and from each other.

[0112] The switch **S54** is connected between the input terminal **134** and the output terminal **130**. Specifically, the switch **S54** has a terminal connected to the input terminal **134**, and a terminal connected to the output terminal **130**. With the connection arrangement mentioned above, the switch **S54** is switched between ON and OFF based on the control signal **S3** to allow switching between connection and disconnection of the input terminal **134** and the output terminal **130** to and from each other.

[0113] The switches **S51** and **S52** correspond to at least one switch included in the output switching circuit **30**.

[0114] The switches **S51** to **S54** are controlled to be exclusively ON. That is, only one of the switches **S51** to **S54** is turned ON, with the remainder of the switches **S51** to **S54** being turned OFF. This allows the output switching circuit **30** to output one voltage selected from among the voltages **V1** to **V4**.

[0115] The configuration of the output switching circuit **30** in FIG. 3 is illustrative and not intended to be limiting. In particular, the switches **S51** to **S54** may have any configuration that allows at least

one of the four input terminals **131** to **134** to be selectively connected to the output terminal **130**. In one alternative example, the output switching circuit **30** may further include a switch connected between: the switches **S51** to **S53**; and the switch **S54** and the output terminal **130**. In another alternative example, the output switching circuit **30** may further include a switch connected between: the switches **S51** and **S52**; and the switches **S53** and **S54** and the output terminal **130**. [0116] According to an exemplary aspect, when two discrete levels of voltage are to be supplied from the switched-capacitor circuit **20**, the output switching circuit **30** may simply include at least two of the switches **S51** to **S54**.

2.3 Circuit Configuration of Pre-Regulator Circuit **10**

[0117] The configuration of the pre-regulator circuit **10** will now be described with reference to FIG. **3**. As illustrated in FIG. **3**, the pre-regulator circuit **10** includes an input terminal **110**, output terminals **111** to **114**, inductor connection terminals **115** and **116**, switches **S61** to **S63**, **S71** and **S72**, a power inductor **L71**, and capacitors **C61** to **C64**.

[0118] The input terminal **110** is an input terminal for a DC voltage. That is, the input terminal **110** is a terminal for receiving an input voltage from the DC power source **50**.

[0119] The output terminal **111** is an output terminal for the voltage **V4**. That is, the output terminal **111** is a terminal for supplying the voltage **V4** to the switched-capacitor circuit **20**. The output terminal **111** is connected to the node **N4** of the switched-capacitor circuit **20**.

[0120] The output terminal **112** is an output terminal for the voltage **V3**. That is, the output terminal **112** is a terminal for supplying the voltage **V3** to the switched-capacitor circuit **20**. The output terminal **112** is connected to the node **N3** of the switched-capacitor circuit **20**.

[0121] The output terminal **113** is an output terminal for the voltage **V2**. That is, the output terminal **113** is a terminal for supplying the voltage **V2** to the switched-capacitor circuit **20**. The output terminal **113** is connected to the node **N2** of the switched-capacitor circuit **20**.

[0122] The output terminal **114** is an output terminal for the voltage **V1**. That is, the output terminal **114** is a terminal for supplying the voltage **V1** to the switched-capacitor circuit **20**. The output terminal **114** is connected to the node **N1** of the switched-capacitor circuit **20**.

[0123] The inductor connection terminal **115** is connected to one end of the power inductor **L71**. The inductor connection terminal **116** is connected to the other end of the power inductor **L71**. For purposes of the exemplary aspects of the present disclosure, the term “one end” can be considered or referred to as a first end. Similarly, the term “other end” can be considered or referred to as a second end.

[0124] The switch **S71** is an example of an eleventh switch, and is connected between the input terminal **110** and the one end of the power inductor **L71**. Specifically, the switch **S71** has a terminal connected to the input terminal **110**, and a terminal connected to the one end of the power inductor **L71** via the inductor connection terminal **115**. With the connection arrangement mentioned above, the switch **S71** is switched between ON and OFF based on a control signal **S1** to allow switching between connection and disconnection of the input terminal **110** and the one end of the power inductor **L71** to and from each other.

[0125] The switch **S72** is an example of a twelfth switch, and is connected between the one end of the power inductor **L71** and ground. Specifically, the switch **S72** has a terminal connected to the one end of the power inductor **L71** via the inductor connection terminal **115**, and a terminal connected to ground. With the connection arrangement mentioned above, the switch **S72** can be switched between ON and OFF based on the control signal **S1** to allow switching between connection and disconnection of the one end of the power inductor **L71** and ground to and from each other.

[0126] The switch **S61** is connected between the other end of the power inductor **L71** and the output terminal **111**. Specifically, the switch **S61** has a terminal connected to the other end of the power inductor **L71** via the inductor connection terminal **116**, and a terminal connected to the output terminal **111**. With the connection arrangement mentioned above, the switch **S61** can be

switched between ON and OFF based on the control signal **S1** to allow switching between connection and disconnection of the other one end of the power inductor **L71** and the output terminal **111** to and from each other.

[0127] The switch **S62** is connected between the other end of the power inductor **L71** and the output terminal **112**. Specifically, the switch **S62** has a terminal connected to the other end of the power inductor **L71** via the inductor connection terminal **116**, and a terminal connected to the output terminal **112**. With the connection arrangement mentioned above, the switch **S62** can be switched between ON and OFF based on the control signal **S1** to allow switching between connection and disconnection of the other one end of the power inductor **L71** and the output terminal **112** to and from each other.

[0128] The switch **S63** is connected between the other end of the power inductor **L71** and the output terminal **113**. Specifically, the switch **S63** has a terminal connected to the other end of the power inductor **L71** via the inductor connection terminal **116**, and a terminal connected to the output terminal **113**. With the connection arrangement mentioned above, the switch **S63** can be switched between ON and OFF based on the control signal **S1** to allow switching between connection and disconnection of the other one end of the power inductor **L71** and the output terminal **113** to and from each other.

[0129] The switches **S71** and **S72** correspond to at least one switch included in the pre-regulator circuit **10**.

[0130] One of two electrodes of the capacitor **C61** is connected to the switch **S61** and the output terminal **111**. The other of the two electrodes of the capacitor **C61** is connected to the switch **S62**, the output terminal **112**, and one of two electrodes of the capacitor **C62**. The capacitor **C61** can be configured to function as a smoothing capacitor in an exemplary aspect.

[0131] The one of the two electrodes of the capacitor **C62** is connected to the switch **S62**, the output terminal **112**, and the other of the two electrodes of the capacitor **C61**. The other of the two electrodes of the capacitor **C62** is connected to a path that connects the switch **S63**, the output terminal **113**, and one of two electrodes of the capacitor **C63**.

[0132] The one of the two electrodes of the capacitor **C63** is connected to the switch **S63**, the output terminal **113**, and the other of the two electrodes of the capacitor **C62**. The other of the two electrodes of the capacitor **C63** is connected to the output terminal **114** and one of two electrodes of the capacitor **C64**.

[0133] The one of the two electrodes of the capacitor **C64** is connected to the output terminal **114** and the other of the two electrodes of the capacitor **C63**. The other of the two electrodes of the capacitor **C64** is connected to ground.

[0134] The switches **S61** to **S63** are controlled to be exclusively ON. That is, only one of the switches **S61** to **S63** is turned ON, with the remainder of the switches **S61** to **S63** being turned OFF. Turning ON only one of the switches **S61** to **S63** as described above allows the pre-regulator circuit **10** to vary the supply voltage for the switched-capacitor circuit **20** between voltage levels corresponding to the voltages **V2** to **V4**.

[0135] The pre-regulator circuit **10** is configured, as described above, to supply charge to the switched-capacitor circuit **20** via at least one of the output terminals **111** to **113**.

[0136] According to an exemplary aspect, when an input voltage is to be converted into a single first voltage, the pre-regulator circuit **10** may only need to include at least the switches **S71** and **S72**, and the power inductor **L71**.

2.4 Circuit Configuration of Filter Circuit **40**

[0137] The configuration of the filter circuit **40** will now be described with reference to FIG. 3. As illustrated in FIG. 4, the filter circuit **40** includes inductors **L1** and **L2**, a capacitor **C1**, a switch **SW1**, the input terminal **140**, and an output terminal **141**.

[0138] The input terminal **140** is connected to the output terminal **130** of the output switching circuit **30**. The input terminal **140** is a terminal for receiving a voltage selected by the output

switching circuit **30** from among a plurality of discrete voltages.

[0139] The output terminal **141** is an external connection terminal of the tracker circuit **1**. The output terminal **141** is connected at a location outside the tracker circuit **1** to the power amplifier **2**. The output terminal **141** is a terminal for supplying, to the power amplifier **2**, a plurality of discrete voltages VA that have passed through the filter circuit **40**.

[0140] The inductor L1 is connected between the input terminal **140** and the output terminal **141**. That is, the inductor L1 is disposed in series with a path connecting the input terminal **140** and the output terminal **141**. Specifically, one end of the inductor L1 is connected to the input terminal **140**, and the other end of the inductor L1 is connected to the output terminal **141**.

[0141] The inductor L2 is connected between a path connecting the inductor L1 and the output terminal **141**, and ground. That is, the inductor L2 is connected in shunt with the path connecting the input terminal **140** and the output terminal **141**. Specifically, one end of the inductor L2 is connected to a node N42 on the path connecting the inductor L1 and the output terminal **141**, and the other end of the inductor L2 is connected to ground via the capacitor C1. The inductor L2 may be connected between the capacitor C1 and ground, and need not necessarily be included in the filter circuit **40**.

[0142] The capacitor C1 is connected between the inductor L2 and ground. That is, the capacitor C1 is connected in shunt with the path connecting the input terminal **140** and the output terminal **141**. Specifically, one end of the capacitor C1 is connected to the inductor L2, and the other end of the capacitor C1 is connected to ground.

[0143] The switch SW1 is connected between the input terminal **140** and the output terminal **141**, without the inductor L1 being interposed between the switch SW1 and each of these terminals. That is, the switch SW1 is disposed in series with a path located between the input terminal **140** and the output terminal **141** and bypassing the inductor L1. Specifically, one terminal of the switch SW1 is connected to a node N41, which is located on a path connecting the input terminal **140** and the inductor L1, and the other terminal of the switch SW1 is connected to a node N43, which is located on a path connecting the inductor L1 and the output terminal **141**.

[0144] Although the switch SW1 and the inductor L1 are connected in parallel with each other in FIG. 3, another circuit element may be inserted into the path where the switch SW1 is located, and/or the path where the inductor L1 is located. For example, an inductor may be connected between the switch SW1 and the node N43, and/or between the switch SW1 and the node N41.

[0145] In FIG. 3, the node N43, which is connected with the other terminal of the switch SW1, is located between the node N42, which is connected with the inductor L2, and the output terminal **141**. The positional relationship between the nodes N42 and N43, however, is not limited to this. In another example, the node N42 may be located between the node N43 and the output terminal **141**. In still another example, the node N42 may be located at the same position as the node N43.

[0146] The switch SW1 connected as described above is switched between ON and OFF based on a control signal S4. This allows the filter circuit **40** to switch between ON and OFF of a band-reject filter used for removing noise from a plurality of discrete voltages.

[0147] The ON/OFF of the band-reject filter described above can be controlled based on, for example, the channel band width (i.e., the modulation band width) of the radio frequency signal RFA. According to an exemplary aspect, when the power amplifier **2** is configured to amplify transmission signals of a plurality of bands, ON/OFF of the switch SW1 may be controlled based on the band of the transmission signal to be amplified by the power amplifier **2**. The manner of control of the band-reject filter is not limited to the manner mentioned above.

2.5 Circuit Configuration of Digital Control Circuit **60**

[0148] The circuit configuration of the digital control circuit **60** will now be described. As illustrated in FIG. 4, the digital control circuit **60** includes a first controller **61**, a second controller **62**, capacitors C81 and C82, and control terminals **601** to **604**.

[0149] The first controller **61** can be configured to generate the control signals S1, S2, and S4 by

processing source-synchronous digital control signals received from the RFIC 5 via the control terminals **601** and **602**. The control signal **S1** is a signal for controlling the ON/OFF of the switches **S61** to **S63**, **S71**, and **S72** included in the pre-regulator circuit **10**. The control signal **S2** is a signal for controlling the ON/OFF of the switches **S11** to **S14**, **S21** to **S24**, **S31** to **S34**, and **S41** to **S44** included in the switched-capacitor circuit **20**. The control signal **S4** is a signal for controlling the ON/OFF of the switch **SW1** included in the filter circuit **40**.

[0150] The digital control signals to be processed by the first controller **61** are not limited to source-synchronous digital control signals. For example, the first controller **61** may process clock-embedded digital control signals. The first controller **61** may generate a control signal for controlling the output switching circuit **30**.

[0151] According to the embodiments, a single set of a clock signal and a data signal is used as digital control signals for controlling the pre-regulator circuit **10**, the switched-capacitor circuit **20**, and the filter circuit **40**. This configuration, however, is not intended to be limiting. For example, sets of a clock signal and a data signal may be used individually as digital control signals for controlling the pre-regulator circuit **10**, the switched-capacitor circuit **20**, and the filter circuit **40**.

[0152] The second controller **62** generates the control signal **S3** by processing digital control logic/line (DCL) signals (**DCL1** and **DCL2**) received from the RFIC 5 via the control terminals **603** and **604**. The DCL signals (**DCL1** and **DCL2**) are generated by the RFIC 5 based on, for example, the envelope signal of a radio frequency signal. The control signal **S3** is a signal for controlling the ON/OFF of the switches **S51** to **S54** included in the output switching circuit **30**.

[0153] The DCL signals (**DCL1** and **DCL2**) are each a 1-bit signal. The voltages **V1** to **V4** are each represented by a combination of two 1-bit signals. For example, **V1**, **V2**, **V3**, and **V4** are represented by “00”, “01”, “10”, and “11”, respectively. Gray code may be used to represent a voltage level.

[0154] The capacitor **C81** is connected between the first controller **61** and ground. For example, the capacitor **C81** is connected between ground and a power supply line that supplies power to the first controller **61**, and can be configured to function as a bypass capacitor in an exemplary aspect. The capacitor **C82** is connected between the second controller **62** and ground.

[0155] According to the embodiments, two DCL signals are used to control the output switching circuit **30**. The number of DCL signals, however, is not limited to two. For example, one DCL signal, or any number of DCL signals greater than or equal to three may be used in accordance with the number of voltage levels selectable by each output switching circuit **30**. The digital control signal to be used for controlling the output switching circuit **30** is not limited to a DCL signal.

3 Implementation Example of Tracker Module **1A** according to Example

[0156] A tracker module **1A** will now be described with reference to FIGS. 5A to 5C as an implementation example of the tracker circuit **1** configured as described above.

[0157] According to the implementation example and modifications thereof described later, the power inductor **L71** included in the pre-regulator circuit **10** is not disposed at a module laminate **90**. This configuration, however, is not intended to be limiting. That is, the power inductor **L71** may be disposed at the module laminate **90**.

[0158] FIG. 5A is a plan view of the tracker module **1A** according to Example. FIG. 5B is a plan view of the tracker module **1A** according to Example. FIG. 5A is a view of a major face **90a** of the module laminate **90** as seen from the positive side of the z-axis. FIG. 5B is a see-through view of a major face **90b** of the module laminate **90** as seen from the positive side of the z-axis. FIG. 5C is a cross-sectional view of the tracker module **1A** according to Example. The cross-section of the tracker module **1A** in FIG. 5C is taken along line VC-VC in each of FIGS. 5A and 5B.

[0159] In FIGS. 5A to 5C, the illustration of some of wiring lines connecting a plurality of circuit components disposed at the module laminate **90** is omitted. In FIGS. 5A to 5C, the illustration of a resin member **91** covering the plurality of circuit components and a shield electrode layer covering the surface of the resin member **91** is omitted. The resin member **91** and the shield electrode layer

need not necessarily be provided. In FIG. 5A, hatched blocks represent optional circuit components that may be omitted in certain exemplary aspects.

[0160] As illustrated in FIG. 5A, the tracker module **1A** includes the module laminate **90**, and an integrated circuit **80**.

[0161] The module laminate **90** has the major faces **90a** and **90b** that are opposite to each other. A ground plane or other components are provided in the module laminate **90** and on the major face **90a**. Although the module laminate **90** is depicted in FIGS. 5A and 5B as having a rectangular shape in plan view, the shape of the module laminate **90** is not limited to a rectangular shape.

[0162] Suitable examples of the module laminate **90** may include, but are not limited to: a low temperature co-fired ceramics (LTCC) substrate or a high temperature co-fired ceramics (HTCC) substrate that has a multilayer structure of a plurality of dielectric layers; a component-embedded substrate; a substrate with a redistribution layer (RDL); and a printed circuit board.

[0163] The integrated circuit **80** is disposed at the major face **90a** of the module laminate **90**. The integrated circuit **80** includes a PR switch portion **10A**, an SC switch portion **20A**, an OS switch portion **30A**, an FL switch portion **40A**, and a digital control portion **60A**. The PR switch portion **10A** includes the switches **S61** to **S63**, **S71**, and **S72** of the pre-regulator circuit **10**. The SC switch portion **20A** includes the switches **S11** to **S14**, **S21** to **S24**, **S31** to **S34**, and **S41** to **S44** of the switched-capacitor circuit **20**. The OS switch portion **30A** includes the switches **S51** to **S54** of the output switching circuit **30**. The FL switch portion **40A** includes the switch **SW1** of the filter circuit **40**. The digital control portion **60A** includes the digital control circuit **60**.

[0164] Although the integrated circuit **80** is depicted in FIG. 5A as having a rectangular shape in plan view of the module laminate **90**, the shape of the integrated circuit **80** is not limited to a rectangular shape.

[0165] The integrated circuit **80** is implemented by using, for example, complementary metal oxide semiconductor (CMOS). Specifically, the integrated circuit **80** may be manufactured by a silicon on insulator (SOI) process. The integrated circuit **80** is not limited to CMOS.

[0166] As illustrated in FIGS. 5A and 5C, with the module laminate **90** seen in plan view (seen in the z-axis direction), the digital control portion **60A** is disposed between the PR switch portion **10A** and the SC switch portion **20A**. In other words, in plan view of the module laminate **90**, the digital control circuit **60** is disposed between the plurality of switches included in the pre-regulator circuit **10**, and the plurality of switches included in the switched-capacitor circuit **20**.

[0167] According to the above-mentioned configuration, the digital control circuit **60**, which generates a relatively small amount of heat, is disposed between the PR switch portion **10A** and the SC switch portion **20A**, which are sources of heat generation. This configuration distributes and dissipates the heat generated by the PR switch portion **10A** and the SC switch portion **20A**. Therefore, efficient heat dissipation with reduced localized temperature rise is achieved. This configuration reduces power consumption of the tracker module **1A**, and reduces efficiency degradation of the tracker module **1A**.

[0168] Further, a sufficient distance can be provided between the PR switch portion **10A** and the SC switch portion **20A**, which perform switching action. This configuration reduces noise interference caused by the switching action.

[0169] According to the embodiments, each of the switches forming the switch portions at least includes two terminals, and one or more semiconductor elements connected between the two terminals. The one or more semiconductor elements are disposed (stacked) in series with a path connecting the two terminals, and are connected in series with each other. In the switch, when the one or more semiconductor elements form a series-connected circuit, a plurality of such series-connected circuits may be connected in parallel between the two terminals. In one example, each of the one or more semiconductor elements is a field effect transistor (FET) including a source electrode, a drain electrode, and a gate electrode. In another example, each of the one or more semiconductor elements may be a bipolar transistor or a diode.

[0170] FIG. 5D is a plan view of a portion of the integrated circuit **80** according to Example. FIG. 5D illustrates the integrated circuit **80** as seen from the positive side of the z-axis. As illustrated in FIG. 5D, in each of the switches forming the switch portions, one or more series-connected semiconductor elements are stacked in a direction parallel to the xy-plane, and a plurality of series-connected circuits each including one or more semiconductor elements are disposed in parallel in the direction parallel to the xy-plane. Alternatively, in each of the switches forming the switch portions, one or more series-connected semiconductor elements may be stacked in the z-axis direction, and a plurality of series-connected circuits each including one or more semiconductor elements may be disposed in parallel in the direction parallel to the xy-plane.

[0171] As used in the description of the embodiments, expressions such as “circuit A or component A is disposed between a plurality of first switches and a plurality of second switches” can indicate that at least one of a plurality of line segments each connecting a given point within one of the plurality of first switches and a given point within one of the plurality of second switches passes through the circuit A or the component A.

[0172] This is explained below with reference to FIG. 5D. The expressions such as “circuit A or component A is disposed between a plurality of first switches and a plurality of second switches” can indicate, for example, that the digital control portion **60A** (at any one of Position “a”, Position “b”, Position “c”, and Position “d”) is disposed between the switches **S61**, **S62**, and **S63** (e.g., a plurality of first switches) of the PR switch portion **10A**, and the switches **S11** to **S14**, **S21** to **S24**, **S31** to **S34**, and **S41** to **S44** (e.g., a plurality of second switches) of the SC switch portion **20A**.

[0173] Specifically, in one example, in FIG. 5D, a line segment **La** connects a given point within the switch **S61**, which is one of the first switches of the PR switch portion **10A**, and a given point within the switch **S12**, which is one of the second switches of the SC switch portion **20A**, and the line segment **La** passes through the digital control portion (at Position “a”). In another example, a line segment **Lb** connects a given point within the switch **S62**, which is one of the first switches of the PR switch portion **10A**, and a given point within the switch **S21**, which is one of the second switches of the SC switch portion **20A**, and the line segment **Lb** passes through the digital control portion (at Position “b”). In another example, a line segment **Lc** connects a given point within the switch **S63**, which is one of the first switches of the PR switch portion **10A**, and a given point within the switch **S41**, which is one of the second switches of the SC switch portion **20A**, and the line segment **Lc** passes through the digital control portion (at each of Position “c” and Position “d”).

[0174] The tracker module **1A** further includes the following components in addition to the module laminate **90** and the integrated circuit **80**: the capacitors **C61** to **C64** included in the pre-regulator circuit **10**; the capacitors **C11** to **C16** and the capacitors **C10** to **C40** included in the switched-capacitor circuit **20**; the inductors **L1** and **L2** and the capacitor **C1** included in the filter circuit **40**; the resin member **91**; and a plurality of external connection terminals **150**.

[0175] The integrated circuit **80**, the capacitors **C61** to **C64**, the capacitors **C11** to **C16**, the capacitors **C10** to **C40**, the inductors **L1** and **L2**, the capacitor **C1**, and the resin member **91** are disposed on the major face **90a**.

[0176] The plurality of external connection terminals **150** are disposed on the major face **90b**. At least one of the plurality of external connection terminals **150** is connected to the output terminal **141** illustrated in FIG. 3. The plurality of external connection terminals **150** are electrically connected, via connections such as via-conductors provided in the module laminate **90**, to the plurality of electronic components disposed on the major face **90a**. The plurality of external connection terminals **150** to be used may be, but are not limited to, copper electrodes. For example, the plurality of external connection terminals **150** to be used may be solder electrodes.

[0177] The resin member **91** covers the major face **90a**, and at least a subset of the plurality of electronic components disposed on the major face **90a**. The resin member **91** serves to ensure reliability, such as mechanical strength and moisture resistance, of the plurality of electronic

components disposed on the major face **90a**. The resin member **91** need not necessarily be included in the tracker module **1A**.

[0178] The capacitors **C61** to **C64**, the capacitors **C11** to **C16**, and the capacitors **C10** to **C40** are implemented as chip capacitors according to an exemplary aspect. Moreover, a chip capacitor can refer to a surface mount device (SMD) forming a capacitor. However, it is noted that the plurality of capacitors mentioned above need not necessarily be implemented as chip capacitors. For example, a subset or all of the plurality of capacitors may be included in an integrated passive device (IPD) or may be included in the integrated circuit **80**.

[0179] The inductors **L1** and **L2** are implemented as chip inductors according to an exemplary aspect. A chip inductor can refer to an SMD forming an inductor. The inductors **L1** and **L2** need not necessarily be implemented as chip inductors. For example, a subset or all of the inductors **L1** and **L2** may be included in an IPD.

[0180] The plurality of capacitors and inductors disposed on the major face **90a** as described above are grouped for each individual circuit and disposed around the integrated circuit **80**.

[0181] Specifically, a group of the capacitors **C61** to **C64** included in the pre-regulator circuit **10** is disposed in a region on the major face **90a** located between the following two straight lines in plan view of the module laminate **90**: a straight line extending along the left side of the integrated circuit **80**; and a straight line extending along the left side of the module laminate **90**. The group of circuit components included in the pre-regulator circuit **10** is thus disposed near the PR switch portion **10A** located within the integrated circuit **80**.

[0182] A group of the capacitors **C11** to **C16** and the capacitors **C10** to **C40** included in the switched-capacitor circuit **20** is disposed in the following two regions on the major face **90a** in plan view of the module laminate **90**: a region located between a straight line extending along the top side of the integrated circuit **80** and a straight line extending along the top side of the module laminate **90**; and a region located between a straight line extending along the right side of the integrated circuit **80** and a straight line extending along the right side of the module laminate **90**. The group of circuit components included in the switched-capacitor circuit **20** is thus disposed near the SC switch portion **20A** located within the integrated circuit **80**. That is, the SC switch portion **20A** is disposed closer to the switched-capacitor circuit **20** than is each of the PR switch portion **10A** and the OS switch portion **30A**.

[0183] A group of the inductors **L1** and **L2** and the capacitor **C1** included in the filter circuit **40** is disposed in a region on the major face **90a** located between the following two lines in plan view of the module laminate **90**: a straight line extending along the bottom side of the integrated circuit **80**; and a straight line extending along the bottom side of the module laminate **90**. The group of circuit components included in the filter circuit **40** is thus disposed near the FL switch portion **40A** located within the integrated circuit **80**. That is, the FL switch portion **40A** is disposed closer to the inductors **L1** and **L2** and the capacitor **C1** of the filter circuit **40** than is each of the PR switch portion **10A** and the SC switch portion **20A**.

[0184] The configuration of the tracker module **1A** in FIGS. **5A** to **5C** is illustrative, and not intended to be limiting. For example, as for the capacitors and the inductors disposed on the major face **90a**, a subset of these components may be provided in the module laminate **90**. In another example, as for the capacitors and the inductors disposed on the major face **90a**, a subset of these components need not necessarily be included in the tracker module **1A** and need not necessarily be disposed at the module laminate **90**.

4 Implementation Example of Tracker Module **1B** according to Modification 1

[0185] A tracker module **1B** according to Modification 1 will now be described with reference to FIGS. **6A** and **6B**.

[0186] FIG. **6A** is a plan view of the tracker module **1B** according to Modification 1. FIG. **6A** is a view of the major face **90a** of the module laminate **90** as seen from the positive side of the z-axis. FIG. **6B** is a cross-sectional view of the tracker module **1B** according to Modification 1. The cross-

section of the tracker module **1B** in FIG. **6B** is taken along line **VIB-VIB** in FIG. **6A**.

[0187] As illustrated in FIG. **6A**, the tracker module **1B** includes the module laminate **90**, and integrated circuits **81B**, **82B**, **83B**, **84B**, and **86B**. The tracker module **1B** according to Modification 1 differs from the tracker module **1A** according to Example in the presence of a plurality of integrated circuits instead of a single integrated circuit. With regard to the tracker module **1B** according to Modification 1, its features identical to those of the tracker module **1A** according to Example will not be described in further detail, and the following description will focus mainly on differences from the tracker module **1A**.

[0188] Each of the integrated circuits **81B**, **82B**, **83B**, **84B**, and **86B** is disposed at the major face **90a** of the module laminate **90**.

[0189] The integrated circuit **81B** is an example of a first integrated circuit. The integrated circuit **81B** has the PR switch portion **10A**, and includes the switches **S61** to **S63**, and **S71** and **S72**. The integrated circuit **82B** is an example of a second integrated circuit. The integrated circuit **82B** has the SC switch portion **20A**, and includes the switches **S11** to **S14**, **S21** to **S24**, **S31** to **S34**, and **S41** to **S44**. The integrated circuit **83B** is an example of a third integrated circuit. The integrated circuit **83B** has the OS switch portion **30A** and includes the switches **S51** to **S54**. The integrated circuit **84B** has the FL switch portion **40A** and includes the switch **SW1**. The integrated circuit **86B** includes the digital control circuit **60**.

[0190] Although the integrated circuits **81B**, **82B**, **83B**, **84B**, and **86B** are depicted in FIG. **6A** as each having a rectangular shape in plan view of the module laminate **90**, the shape of these integrated circuits is not limited to a rectangular shape.

[0191] The integrated circuits **81B**, **82B**, **83B**, **84B**, and **86B** are each implemented by using, for example, CMOS. Specifically, each of these integrated circuits may be manufactured by a SOI process. The integrated circuit **80** is not limited to CMOS.

[0192] As illustrated in FIGS. **6A** and **6B**, with the module laminate **90** seen in plan view (seen in the z-axis direction), the integrated circuit **86B** is disposed between the integrated circuit **81B** and the integrated circuit **82B**. In other words, in plan view of the module laminate **90**, the digital control circuit **60** is disposed between the plurality of switches included in the pre-regulator circuit **10** and the plurality of switches included in the switched-capacitor circuit **20**.

[0193] According to the above-mentioned configuration, the integrated circuit **86B** (digital control circuit **60**), which generates a relatively small amount of heat, is disposed between the integrated circuit **81B** (PR switch portion) and the integrated circuit **82B** (SC switch portion), which are sources of heat generation. This configuration distributes and dissipates the heat generated by the PR switch portion and the SC switch portion. Therefore, efficient heat dissipation with reduced localized temperature rise is achieved. This configuration reduces efficiency degradation of the tracker module **1B**.

[0194] Further, a sufficient distance can be provided between the PR switch portion and the SC switch portion, which perform switching action. This configuration reduces noise interference caused by the switching action.

[0195] The tracker module **1B** includes the following components in addition to the module laminate **90** and the integrated circuit **80**: the capacitors **C61** to **C64** included in the pre-regulator circuit **10**; the capacitors **C11** to **C16** and the capacitors **C10** to **C40** included in the switched-capacitor circuit **20**; the inductors **L1** and **L2** and the capacitor **C1** included in the filter circuit **40**; the resin member **91**; and the plurality of external connection terminals **150**.

[0196] The configuration of the tracker module **1B** in FIGS. **6A** and **6B** is illustrative, and not intended to be limiting. For example, as for the capacitors and the inductors disposed on the major face **90a**, a subset of these components may be provided in the module laminate **90**. In another example, as for the capacitors and the inductors disposed on the major face **90a**, a subset of these components need not necessarily be included in the tracker module **1B** and need not necessarily be disposed at the module laminate **90**.

[0197] The FL switch portion may be included in one of the integrated circuits **81B** and **82B**.

5 Implementation Example of Tracker Module **1C** according to Modification 2

[0198] A tracker module **1C** according to Modification 2 will now be described with reference to FIG. 7.

[0199] FIG. 7 is a plan view of the tracker module **1C** according to Modification 2. FIG. 7 is a view of the major face **90a** of the module laminate **90** as seen from the positive side of the z-axis.

[0200] As illustrated in FIG. 7, the tracker module **1C** includes the module laminate **90**, and an integrated circuit **80C**. The tracker module **1C** according to Modification 2 differs from the tracker module **1A** according to Example in the positioning of the switch portions within the integrated circuit **80C**. With regard to the tracker module **1C** according to Modification 2, its features identical to those of the tracker module **1A** according to Example will not be described in further detail, and the following description will focus mainly on differences from the tracker module **1A**.

[0201] The integrated circuit **80C** is disposed at the major face **90a** of the module laminate **90**. The integrated circuit **80C** includes the PR switch portion **10A**, the SC switch portion **20A**, the OS switch portion **30A**, the FL switch portion **40A**, and the digital control portion **60A**. The PR switch portion **10A** includes the switches **S61** to **S63**, **S71**, and **S72**. The SC switch portion **20A** includes the switches **S11** to **S14**, **S21** to **S24**, **S31** to **S34**, and **S41** to **S44**. The OS switch portion **30A** includes the switches **S51** to **S54**. The FL switch portion **40A** includes the switch **SW1**. The digital control portion **60A** includes the digital control circuit **60**.

[0202] Although the integrated circuit **80C** is depicted in FIG. 7 as having a rectangular shape in plan view of the module laminate **90**, the shape of the integrated circuit **80C** is not limited to a rectangular shape.

[0203] As illustrated in FIG. 7, with the module laminate **90** seen in plan view (seen in the z-axis direction), the digital control portion **60A** is disposed between the PR switch portion **10A** and the SC switch portion **20A**. In other words, in plan view of the module laminate **90**, the digital control circuit **60** is disposed between the plurality of switches included in the pre-regulator circuit **10** and the plurality of switches included in the switched-capacitor circuit **20**.

[0204] With the module laminate **90** seen in plan view (seen in the z-axis direction), the PR switch portion **10A**, the SC switch portion **20A**, and the OS switch portion **30A** are disposed in a peripheral region within the integrated circuit **80C**.

[0205] As used herein, expressions such as “circuit A or component A is disposed in a peripheral region within the integrated circuit” can indicate that no other circuit component is disposed between one peripheral side of the integrated circuit and the circuit A or component A.

[0206] The above-mentioned configuration shortens the heat dissipation path from the PR switch portion **10A** and the SC switch portion **20A** to an external circuit connected to the tracker module **1C**, and consequently improves heat dissipation from the tracker module **1C**.

[0207] The above-mentioned configuration also shortens the connection wiring between the PR switch portion **10A** and the capacitors **C61** to **C64** included in the pre-regulator circuit **10**, and shortens the connection wiring between the SC switch portion **20A**, and the capacitors **C11** to **C16** and the capacitors **C10** to **C40** included in the switched-capacitor circuit **20**. As such, this configuration reduces signal transmission loss.

[0208] With the module laminate **90** seen in plan view (seen in the z-axis direction), the digital control portion **60A** is disposed more centrally within the integrated circuit **80C** than are the PR switch portion **10A**, the SC switch portion **20A** and the OS switch portion **30A**.

[0209] According to the above-mentioned configuration, a sufficiently large distance can be provided between the PR switch portion **10A** and the SC switch portion **20A**, which are sources of heat generation. This configuration distributes and dissipates the heat generated by the PR switch portion **10A** and the SC switch portion **20A**.

[0210] The above-mentioned configuration also shortens the control signal wiring that connects the digital control portion **60A** with each of the PR switch portion **10A**, the SC switch portion **20A**, and

the OS switch portion **30A**. This enables the digital control circuit **60** to accurately control the switching action of each switch portion.

[0211] The tracker module **1C** includes the following components in addition to the module laminate **90** and the integrated circuit **80C**: the capacitors **C61** to **C64** included in the pre-regulator circuit **10**; the capacitors **C11** to **C16** and the capacitors **C10** to **C40** included in the switched-capacitor circuit **20**; the inductors **L1** and **L2** and the capacitor **C1** included in the filter circuit **40**; the resin member **91**; and the plurality of external connection terminals **150**.

[0212] The configuration of the tracker module **1C** in FIG. **7** is illustrative, and not intended to be limiting. For example, as for the capacitors and the inductors disposed on the major face **90a**, a subset of these components may be provided in the module laminate **90**. In another example, as for the capacitors and the inductors disposed on the major face **90a**, a subset of these components need not necessarily be included in the tracker module **1C** and need not necessarily be disposed at the module laminate **90**.

6 Implementation Example of Tracker Module **1D** according to Modification 3

[0213] A tracker module **1D** according to Modification 3 will now be described with reference to FIG. **8**.

[0214] FIG. **8** is a plan view of the tracker module **1D** according to Modification 3. FIG. **8** is a view of the major face **90a** of the module laminate **90** as seen from the positive side of the z-axis.

[0215] As illustrated in FIG. **8**, the tracker module **1D** includes the module laminate **90**, and an integrated circuit **80D**. The tracker module **1D** according to Modification 3 differs from the tracker module **1A** according to Example in the positioning of the switch portions within the integrated circuit **80D**. With regard to the tracker module **1D** according to Modification 3, its features identical to those of the tracker module **1A** according to Example will not be described in further detail, and the following description will focus mainly on differences from the tracker module **1A**.

[0216] The integrated circuit **80D** is disposed at the major face **90a** of the module laminate **90**. The integrated circuit **80D** includes the PR switch portion **10A**, the SC switch portion **20A**, the OS switch portion **30A**, the FL switch portion **40A**, and the digital control portion **60A**. The PR switch portion **10A** includes the switches **S61** to **S63**, **S71**, and **S72**. The SC switch portion **20A** includes the switches **S11** to **S14**, **S21** to **S24**, **S31** to **S34**, and **S41** to **S44**. The OS switch portion **30A** includes the switches **S51** to **S54**. The FL switch portion **40A** includes the switch **SW1**. The digital control portion **60A** includes the digital control circuit **60**.

[0217] Although the integrated circuit **80D** is depicted in FIG. **8** as having a rectangular shape in plan view of the module laminate **90**, the shape of the integrated circuit **80D** is not limited to a rectangular shape.

[0218] As illustrated in FIG. **8**, with the module laminate **90** seen in plan view (seen in the z-axis direction), the SC switch portion **20A** is provided in a region formed by two elongated regions extending in mutually intersecting directions (the x-axis direction and the y-axis direction), and the two elongated regions are positioned along two peripheral sides of the integrated circuit **80D** (the right and top sides of the integrated circuit **80D**). In other words, the SC switch portion **20A** is provided in an L-shaped region formed by two elongated regions extending in mutually intersecting directions (the x-axis direction and the y-axis direction) and is disposed in a peripheral region within the integrated circuit **80D**.

[0219] The above-mentioned configuration shortens the connection wiring between the SC switch portion **20A**, and the capacitors **C11** to **C16** and the capacitors **C10** to **C40** included in the switched-capacitor circuit **20**. This configuration also reduces heat generation in the switched-capacitor circuit **20**, and also reduces signal transmission loss.

[0220] The tracker module **1D** includes the following components in addition to the module laminate **90** and the integrated circuit **80D**: the capacitors **C61** to **C64** included in the pre-regulator circuit **10**; the capacitors **C11** to **C16** and the capacitors **C10** to **C40** included in the switched-capacitor circuit **20**; the inductors **L1** and **L2** and the capacitor **C1** included in the filter circuit **40**;

the resin member **91**; and the plurality of external connection terminals **150**.

[0221] In the tracker module **1D**, the at least one capacitor included in the switched-capacitor circuit **20** may be disposed adjacent to the integrated circuit **80D**.

[0222] The above-mentioned configuration further shortens the connection wiring between the SC switch portion **20A**, and the capacitors **C11** to **C16** and the capacitors **C10** to **C40** included in the switched-capacitor circuit **20**. This configuration also reduces heat generation in the switched-capacitor circuit **20**, and also reduces signal transmission loss.

[0223] The configuration of the tracker module **1D** in FIG. **8** is illustrative, and not intended to be limiting. For example, as for the capacitors and the inductors disposed on the major face **90a**, a subset of these components may be provided in the module laminate **90**. In another example, as for the capacitors and the inductors disposed on the major face **90a**, a subset of these components need not necessarily be included in the tracker module **1D** and need not necessarily be disposed at the module laminate **90**.

7 Implementation Example of Tracker Module **1E** according to Modification 4

[0224] A tracker module **1E** according to Modification 4 will now be described with reference to FIGS. **9A** and **9B**.

[0225] FIG. **9A** is a plan view of the tracker module **1E** according to Modification 4. FIG. **9A** is a view of the major face **90a** of the module laminate **90** as seen from the positive side of the z-axis. FIG. **9B** is a cross-sectional view of the tracker module **1E** according to Modification 4. The cross-section of the tracker module **1E** in FIG. **9B** is taken along line IXB-IXB in FIG. **9A**.

[0226] As illustrated in FIG. **9A**, the tracker module **1E** includes the module laminate **90**, and an integrated circuit **80E**. The tracker module **1E** according to Modification 4 differs from the tracker module **1D** according to Modification 3 in the positioning of the OS switch portion **30A** and the digital control portion **60A**. With regard to the tracker module **1E** according to Modification 4, its features identical to those of the tracker module **1D** according to Modification 3 will not be described in further detail, and the following description will focus mainly on differences from the tracker module **1D**.

[0227] The integrated circuit **80E** is disposed at the major face **90a** of the module laminate **90**. The integrated circuit **80E** includes the PR switch portion **10A**, the SC switch portion **20A**, the OS switch portion **30A**, the FL switch portion **40A**, and the digital control portion **60A**. The PR switch portion **10A** includes the switches **S61** to **S63**, **S71**, and **S72**. The SC switch portion **20A** includes the switches **S11** to **S14**, **S21** to **S24**, **S31** to **S34**, and **S41** to **S44**. The OS switch portion **30A** includes the switches **S51** to **S54**. The FL switch portion **40A** includes the switch **SW1**. The digital control portion **60A** includes the digital control circuit **60**.

[0228] As illustrated in FIGS. **9A** and **9B**, with the module laminate **90** seen in plan view (seen in the z-axis direction), the digital control portion **60A** and the OS switch portion **30A** are disposed between the PR switch portion **10A** and the SC switch portion **20A**. In other words, in plan view of the module laminate **90**, the digital control circuit **60**, and the plurality of switches included in the output switching circuit **30** are disposed between the plurality of switches included in the pre-regulator circuit **10** and the plurality of switches included in the switched-capacitor circuit **20**.

[0229] According to the above-mentioned configuration, the digital control circuit **60** and the OS switch portion **30A**, which generate a relatively small amount of heat, are disposed between the PR switch portion **10A** and the SC switch portion **20A**, which are sources of heat generation. This configuration distributes and dissipates the heat generated by the PR switch portion **10A** and the SC switch portion **20A**. Therefore, efficient heat dissipation with reduced localized temperature rise is achieved. This configuration also further reduces efficiency degradation of the tracker module **1E**.

[0230] Further, a sufficiently large distance can be provided between the PR switch portion **10A** and the SC switch portion **20A**, which have a high switching speed relative to the OS switch portion **30A**. This configuration further reduces noise interference caused by the switching action.

[0231] The SC switch portion **20A** and the OS switch portion **30A** are disposed adjacent to each

other. In other words, the plurality of switches included in the switched-capacitor circuit **20**, and the plurality of switches included in the output switching circuit **30** are disposed adjacent to each other. [0232] The above-mentioned configuration connects the SC switch portion **20A** and the OS switch portion **30A** by low-impedance wiring. This configuration improves transient response performance at the rising and falling voltage edges. Further, the low impedance of the above-mentioned wiring serving as a power line reduces power consumption.

[0233] According to the embodiments, expressions such as “a plurality of first switches and a plurality of second switches are disposed adjacent to each other” can indicate that no circuit component is disposed between a given first switch of the plurality of first switches and a given second switch of the plurality of second switches.

[0234] As illustrated in FIG. **9B**, the PR switch portion **10A** and the SC switch portion **20A** are connected via conductor wiring lines **96** and **97** provided at the module laminate **90**. In other words, the plurality of switches included in the switched-capacitor circuit **20** and the plurality of switches included in the pre-regulator circuit **10** are connected to each other via an electrically conductive member provided at the module laminate **90**.

[0235] According to the above-mentioned configuration, the wiring for connecting the PR switch portion **10A** and the SC switch portion **20A** can be provided at the module laminate **90**. This configuration improves the freedom of layout of the wiring, and achieves low impedance. Therefore, the signal transmission loss between the PR switch portion **10A** and the SC switch portion **20A** is reduced.

[0236] The tracker module **1E** includes the following components in addition to the module laminate **90** and the integrated circuit **80E**: the capacitors **C61** to **C64** included in the pre-regulator circuit **10**; the capacitors **C11** to **C16** and the capacitors **C10** to **C40** included in the switched-capacitor circuit **20**; the inductors **L1** and **L2** and the capacitor **C1** included in the filter circuit **40**; the resin member **91**; and the plurality of external connection terminals **150**.

[0237] In the tracker module **1E**, the at least one capacitor included in the switched-capacitor circuit **20** may be disposed adjacent to the integrated circuit **80E**.

[0238] According to the above-mentioned configuration, the wiring for connecting the PR switch portion **10A** and the SC switch portion **20A** can be provided at the module laminate **90**. This configuration improves the freedom of layout of the wiring and achieve low impedance. Therefore, the signal transmission loss between the pre-regulator circuit **10** and the switched-capacitor circuit **20** is reduced.

[0239] The configuration of the tracker module **1E** in FIGS. **9A** and **9B** is illustrative, and not intended to be limiting. For example, as for the capacitors and the inductors disposed on the major face **90a**, a subset of these components may be provided in the module laminate **90**. In another example, as for the capacitors and the inductors disposed on the major face **90a**, a subset of these components need not necessarily be included in the tracker module **1E** and need not necessarily be disposed at the module laminate **90**.

8 Implementation Example of Tracker Module **1F** according to Modification 5

[0240] A tracker module **1F** according to Modification 5 will now be described with reference to FIG. **10**.

[0241] FIG. **10** is a plan view of the tracker module **1F** according to Modification 5. FIG. **10** is a view of the major face **90a** of the module laminate **90** as seen from the positive side of the z-axis.

[0242] As illustrated in FIG. **10**, the tracker module **1F** includes the module laminate **90**, and an integrated circuit **80F**. The tracker module **1F** according to Modification 5 differs from the tracker module **1A** according to Example in the positioning of the switch portions in the integrated circuit **80F**. With regard to the tracker module **1F** according to Modification 5, its features identical to those of the tracker module **1A** according to Example will not be described in further detail, and the following description will focus mainly on differences from the tracker module **1A**.

[0243] The integrated circuit **80F** is disposed at the major face **90a** of the module laminate **90**. The

integrated circuit **80F** includes the PR switch portion **10A**, the SC switch portion **20A**, the OS switch portion **30A**, the FL switch portion **40A**, and the digital control portion **60A**. The PR switch portion **10A** includes the switches **S61** to **S63**, **S71**, and **S72**. The SC switch portion **20A** includes the switches **S11** to **S14**, **S21** to **S24**, **S31** to **S34**, and **S41** to **S44**. The OS switch portion **30A** includes the switches **S51** to **S54**. The FL switch portion **40A** includes the switch **SW1**. The digital control portion **60A** includes the digital control circuit **60**.

[0244] As illustrated in FIG. **10**, the SC switch portion **20A** is disposed separately in two regions that are spaced apart from each other. In other words, the plurality of switches included in the switched-capacitor circuit **20** are provided separately in a first region and a second region that are spaced apart from each other.

[0245] According to the above-mentioned configuration, the SC switch portion **20A**, which is a source of heat generation, is provided separately in two regions. This configuration distributes and dissipates the heat generated by the SC switch portion **20A**.

[0246] With the module laminate **90** seen in plan view (seen in the z-axis direction), the digital control portion **60A** is disposed between the two separately disposed SC switch portions **20A**.

[0247] According to the above-mentioned configuration, the digital control circuit **60**, which generates a relatively small amount of heat, is disposed between the two SC switch portions **20A**, which are sources of heat generation. This configuration distributes and dissipates the heat generated by the SC switch portions **20A**. Therefore, efficient heat dissipation with reduced localized temperature rise is achieved. This configuration further reduces efficiency degradation of the tracker module **1F**.

[0248] The tracker module **1F** includes the following components in addition to the module laminate **90** and the integrated circuit **80F**: the capacitors **C61** to **C64** included in the pre-regulator circuit **10**; the capacitors **C11** to **C16** and the capacitors **C10** to **C40** included in the switched-capacitor circuit **20**; the inductors **L1** and **L2** and the capacitor **C1** included in the filter circuit **40**; the resin member **91**; and the plurality of external connection terminals **150**.

[0249] The configuration of the tracker module **1F** in FIG. **10** is illustrative, and not intended to be limiting. For example, as for the capacitors and the inductors disposed on the major face **90a**, a subset of these components may be provided in the module laminate **90**. In another example, as for the capacitors and the inductors disposed on the major face **90a**, a subset of these components need not necessarily be included in the tracker module **1F** and need not necessarily be disposed at the module laminate **90**.

9 Implementation Example of Tracker Module **1G** according to Modification 6

[0250] A tracker module **1G** according to Modification 6 will now be described with reference to FIGS. **11A** to **11C**.

[0251] FIG. **11A** is a plan view of the tracker module **1G** according to Modification 6. FIG. **11B** is a plan view of the tracker module **1G** according to Modification 6. FIG. **11A** is a view of the major face **90a** of the module laminate **90** as seen from the positive side of the z-axis. FIG. **11B** is a see-through view of the major face **90b** of the module laminate **90** as seen from the positive side of the z-axis. FIG. **11C** is a cross-sectional view of the tracker module **1G** according to Modification 6. The cross-section of the tracker module **1G** in FIG. **11C** is taken along line XIC-XIC in each of FIGS. **11A** and **11B**.

[0252] As illustrated in FIGS. **11A** and **11B**, the tracker module **1G** includes the module laminate **90**, and integrated circuits **81G**, **82G**, **83G**, **84G**, and **86G**. The tracker module **1G** according to Modification 6 differs from the tracker module **1A** according to Example in that a plurality of integrated circuits are disposed on the major faces **90a** and **90b** of the module laminate **90** in a distributed manner. With regard to the tracker module **1G** according to Modification 6, its features identical to those of the tracker module **1A** according to Example will not be described in further detail, and the following description will focus mainly on differences from the tracker module **1A**.

[0253] Each of the integrated circuits **83G**, **84G**, and **86G** is disposed at the major face **90a** of the

module laminate **90**. Each of the integrated circuits **81G**, **82G**, and **86G** is disposed at the major face **90b** of the module laminate **90**.

[0254] The integrated circuit **81G** is an example of the first integrated circuit. The integrated circuit **81G** has the PR switch portion **10A**, and includes the switches **S61** to **S63**, and **S71** and **S72**. The integrated circuit **82G** is an example of the second integrated circuit. The integrated circuit **82G** has the SC switch portion **20A**, and includes the switches **S11** to **S14**, **S21** to **S24**, **S31** to **S34**, and **S41** to **S44**. The integrated circuit **83G** is an example of the third integrated circuit. The integrated circuit **83G** has the OS switch portion **30A** and includes the switches **S51** to **S54**. The integrated circuit **84G** has the FL switch portion **40A** and includes the switch **SW1**. The integrated circuit **86G** includes the digital control circuit **60**.

[0255] Although the integrated circuits **81G**, **82G**, **83G**, **84G**, and **86G** are depicted in FIGS. **11A** and **11B** as each having a rectangular shape in plan view of the module laminate **90**, the shape of these integrated circuits is not limited to a rectangular shape.

[0256] The tracker module **1G** includes the following components in addition to the module laminate **90** and the plurality of integrated circuits mentioned above: the capacitors **C61** to **C64** included in the pre-regulator circuit **10**; the capacitors **C11** to **C16** and the capacitors **C10** to **C40** included in the switched-capacitor circuit **20**; the inductors **L1** and **L2** and the capacitor **C1** included in the filter circuit **40**; resin members **91** and **92**; and the plurality of external connection terminals **150**.

[0257] In addition to the integrated circuits **83G**, **84G**, and **86G**, the following components are disposed on the major face **90a**: the capacitors **C61** to **C62**, the capacitors **C11**, **C12**, **C14** and **C15**, the capacitors **C10** to **C20**, the inductors **L1** and **L2**, the capacitor **C1**, and the resin member **91**.

[0258] In addition to the integrated circuits **81G**, **82G**, and **86G**, the following components are disposed on the major face **90b**: the capacitors **C63** and **C64**, the capacitors **C13** and **C16**, the capacitors **C30** and **C40**, and the plurality of external connection terminals **150**. At least one of the plurality of external connection terminals **150** is connected to the output terminal **141** illustrated in FIG. **3**. The plurality of external connection terminals **150** are electrically connected, via connections such as via-conductors provided in the module laminate **90**, to the plurality of electronic components disposed on the major faces **90a** and **90b** and are connected to an electrode on a motherboard disposed near the major face **90b**.

[0259] The resin member **91** covers at least a subset of the plurality of electronic components disposed on the major face **90a**. The resin member **91** serves to ensure reliability, such as mechanical strength and moisture resistance, of the plurality of electronic components disposed on the major face **90a**. The resin member **92** covers at least a subset of the plurality of electronic components disposed on the major face **90b**. The resin member **92** serves to ensure reliability, such as mechanical strength and moisture resistance, of the plurality of electronic components disposed on the major face **90b**. The resin members **91** and **92** need not necessarily be included in the tracker module **1G**.

[0260] According to the above-mentioned configuration, the integrated circuit **81G** (PR switch portion) and the integrated circuit **82G** (SC switch portion), which are sources of heat generation, are disposed at the major face **90b**. This configuration shortens the heat dissipation path to the motherboard (e.g., an external circuit), and also improves heat dissipation from the back side of the integrated circuits **81G** and **82G**. Therefore, efficient heat dissipation with reduced temperature rise of the tracker module **1G** is achieved. This configuration also reduces efficiency degradation of the tracker module **1G**.

[0261] The integrated circuit **83G** is disposed at the major face **90a**, and with the module laminate **90** seen in plan view (seen in the z-axis direction), the integrated circuit **82G** and the integrated circuit **83G** at least partially overlap each other.

[0262] The above-mentioned configuration connects the SC switch portion and the OS switch portion by low-impedance wiring. This configuration improves transient response performance at

the rising and falling voltage edges. Further, the low impedance of the above-mentioned wiring serving as a power line reduces power consumption.

[0263] The configuration of the tracker module **1G** in FIGS. **11A** to **11C** is illustrative, and not intended to be limiting. For example, as for the capacitors and the inductors disposed on the major faces **90a** and **90b**, a subset of these components may be provided in the module laminate **90**. In another example, as for the capacitors and the inductors disposed on the major faces **90a** and **90b**, a subset of these components need not necessarily be included in the tracker module **1G** and need not necessarily be disposed at the module laminate **90**.

10 Technical Effects

[0264] As described above, the tracker modules **1A** and **1C** to **1F** according to Example and its modifications include the module laminate **90**, and a single integrated circuit **80** (or **80C**, **80D**, **80E**, or **80F**) disposed at the module laminate **90**. The integrated circuit **80** (or **80C**, **80D**, **80E**, or **80F**) includes: at least one switch included in the pre-regulator circuit **10**, the pre-regulator circuit **10** being configured to convert an input voltage into a first voltage; at least one switch included in the switched-capacitor circuit **20**, the switched-capacitor circuit **20** being configured to generate a plurality of discrete second voltages based on the first voltage; at least one switch included in the output switching circuit **30**, the output switching circuit **30** being configured to selectively output at least one of the plurality of discrete second voltages generated by the switched-capacitor circuit **20**; and the digital control circuit **60** configured to, based on a digital control signal corresponding to an envelope signal, cause the output switching circuit **30** to selectively output at least one of the plurality of discrete second voltages. In plan view of the module laminate **90**, at least one of: the at least one switch included in the output switching circuit **30**; or the digital control circuit **60** is disposed between the at least one switch included in the pre-regulator circuit **10** and the at least one switch included in the switched-capacitor circuit **20**.

[0265] According to the above-mentioned configuration, at least one of the digital control circuit **60** or the OS switch portion **30A**, which generates a relatively small amount of heat, is disposed between the PR switch portion **10A** and the SC switch portion **20A**, which are sources of heat generation. This configuration distributes and dissipates the heat generated by the PR switch portion **10A** and the SC switch portion **20A**. Therefore, efficient heat dissipation with reduced localized temperature rise is achieved. This configuration also reduces efficiency degradation of the tracker modules **1A**, and **1C** to **1F**.

[0266] The tracker modules **1B** and **1G** according to modifications include: the module laminate **90**; the integrated circuit **81B** (or **81G**) disposed at the module laminate **90** and including at least one switch included in the pre-regulator circuit **10**, the pre-regulator circuit **10** being configured to convert an input voltage into a first voltage; the integrated circuit **82B** (or **82G**) disposed at the module laminate **90** and including at least one switch included in the switched-capacitor circuit **20**, the switched-capacitor circuit **20** being configured to generate a plurality of discrete second voltages based on the first voltage; the integrated circuit **83B** (or **83G**) disposed at the module laminate **90** and including at least one switch included in the output switching circuit **30**, the output switching circuit **30** being configured to selectively output at least one of the plurality of discrete second voltages generated by the switched-capacitor circuit **20**; and the digital control circuit **60** configured to, based on a digital control signal corresponding to an envelope signal, cause the output switching circuit **30** to selectively output at least one of the plurality of discrete second voltages. In plan view of the module laminate **90**, at least one of: the at least one switch included in the output switching circuit **30**; or the digital control circuit **60** is disposed between the at least one switch included in the pre-regulator circuit **10** and the at least one switch included in the switched-capacitor circuit **20**.

[0267] According to the above-mentioned configuration, at least one of the digital control circuit **60** or the OS switch portion **30A**, which generates a relatively small amount of heat, is disposed between the PR switch portion **10A** and the SC switch portion **20A**, which are sources of heat

generation. This configuration distributes and dissipates the heat generated by the PR switch portion **10A** and the SC switch portion **20A**. Therefore, efficient heat dissipation with reduced localized temperature rise is achieved. This configuration also reduces efficiency degradation of the tracker modules **1B** and **1G**.

[0268] In one exemplary configuration of the tracker module **1G**, the module laminate **90** has the major faces **90a** and **90b** that are opposite to each other, the tracker module **1G** further includes the external connection terminal **150** disposed at the major face **90b**, and the integrated circuits **81G** and **82G** are disposed at the major face **90b**.

[0269] According to the above-mentioned configuration, the integrated circuit **81G** (PR switch portion) and the integrated circuit **82G** (SC switch portion), which are sources of heat generation, are disposed at the major face **90b**. This configuration shortens the heat dissipation path to the motherboard (e.g., an external circuit), and also improve heat dissipation from the back side of the integrated circuits **81G** and **82G**. Therefore, efficient heat dissipation with reduced temperature rise of the tracker module **1G** is achieved. This configuration also reduces efficiency degradation of the tracker module **1G**.

[0270] In one exemplary configuration of the tracker module **1G**, the integrated circuit **83G** is disposed at the major face **90a**, and in plan view of the module laminate **90**, the integrated circuit **82G** and the integrated circuit **83G** at least partially overlap each other.

[0271] The above-mentioned configuration connects the SC switch portion **20A** and the OS switch portion **30A** by low-impedance wiring. This configuration also improves transient response performance at the rising and falling voltage edges. Further, the low impedance of the above-mentioned wiring serving as a power line reduces power consumption.

[0272] In one exemplary configuration of the tracker modules **1E** and **1F**, the at least one switch included in the switched-capacitor circuit **20**, and the at least one switch included in the output switching circuit **30** are disposed adjacent to each other.

[0273] The above-mentioned configuration connects the SC switch portion **20A** and the OS switch portion **30A** by low-impedance wiring. This configuration also improves transient response performance at the rising and falling voltage edges. Further, the low impedance of the above-mentioned wiring serving as a power line reduces power consumption.

[0274] In one exemplary configuration of the tracker modules **1D** and **1E**, in plan view of the module laminate **90**, the at least one switch included in the output switching circuit **30**, and the digital control circuit **60** are disposed between the at least one switch included in the pre-regulator circuit **10** and the at least one switch included in the switched-capacitor circuit **20**.

[0275] According to the above-mentioned configuration, the digital control circuit **60** and the OS switch portion **30A**, which generate a relatively small amount of heat, are disposed between the PR switch portion **10A** and the SC switch portion **20A**, which are sources of heat generation. This configuration distributes and dissipates the heat generated by the PR switch portion **10A** and the SC switch portion **20A**. Therefore, efficient heat dissipation with reduced localized temperature rise is achieved. This configuration further reduces efficiency degradation of the tracker modules **1D** and **1E**.

[0276] In one exemplary configuration of the tracker module **1C**, in plan view of the module laminate **90**, the at least one switch included in the pre-regulator circuit **10**, the at least one switch included in the switched-capacitor circuit **20**, and the at least one switch included in the output switching circuit **30** are disposed in a peripheral region within the integrated circuit **80C**.

[0277] The above-mentioned configuration shortens the heat dissipation path from the PR switch portion **10A** and the SC switch portion **20A** to an external circuit connected to the tracker module **1C**, and consequently improve heat dissipation from the tracker module **1C**. The above-mentioned configuration also shortens the connection wiring between the PR switch portion **10A** and the capacitors included in the pre-regulator circuit **10**, and to shorten the connection wiring between the SC switch portion **20A** and the capacitors included in the switched-capacitor circuit **20**. This

configuration reduces signal transmission loss.

[0278] In one exemplary configuration of the tracker module **1C**, in plan view of the module laminate **90**, the digital control circuit **60** is disposed more centrally within the integrated circuit **80C** than are the at least one switch included in the pre-regulator circuit **10**, the at least one switch included in the switched-capacitor circuit **20**, and the at least one switch included in the output switching circuit **30**.

[0279] According to the above-mentioned configuration, a sufficient distance can be provided between the PR switch portion **10A** and the SC switch portion **20A**, which are sources of heat generation. This configuration distributes and dissipates the heat generated by the PR switch portion **10A** and the SC switch portion **20A**. The above-mentioned configuration also shortens the control signal wiring that connects the digital control portion **60A** with each of the PR switch portion **10A**, the SC switch portion **20A**, and the OS switch portion **30A**. This configuration enables the digital control circuit **60** to accurately control the switching action of each switch portion.

[0280] In one exemplary configuration of the tracker modules **1D** and **1E**, in plan view of the module laminate **90**, the at least one switch included in the switched-capacitor circuit **20** is provided in a region formed by two elongated regions extending in mutually intersecting directions, and the two elongated regions are positioned along two peripheral sides of the integrated circuit **80D** (or **80E**).

[0281] The above-mentioned configuration shortens the connection wiring between the SC switch portion **20A** and the capacitors included in the switched-capacitor circuit **20**. This configuration reduces heat generation in the switched-capacitor circuit **20**, and also reduces signal transmission loss.

[0282] In one exemplary configuration of the tracker modules **1D** and **1E**, the at least one capacitor included in the switched-capacitor circuit **20** may be disposed adjacent to the integrated circuit **80D** (or **80E**).

[0283] The above-mentioned configuration further shortens the connection wiring between the SC switch portion **20A** and the capacitors included in the switched-capacitor circuit **20**. This configuration reduces heat generation in the switched-capacitor circuit **20**, and also reduces signal transmission loss.

[0284] In one exemplary configuration of the tracker module **1F**, the at least one switch included in the switched-capacitor circuit **20** is provided separately in a first region and a second region that are spaced apart from each other.

[0285] According to the above-mentioned configuration, the SC switch portion **20A**, which is a source of heat generation, is provided separately in two regions. This configuration distributes and dissipates the heat generated by the SC switch portion **20A**.

[0286] In one exemplary configuration of the tracker module **1F**, in plan view of the module laminate **90**, the digital control circuit **60** is disposed between the first region and the second region.

[0287] According to the above-mentioned configuration, the digital control circuit **60**, which generates a relatively small amount of heat, is disposed between the two SC switch portions **20A**, which are sources of heat generation. This configuration distributes and dissipates the heat generated by the SC switch portions **20A**. Therefore, efficient heat dissipation with reduced localized temperature rise is achieved. This configuration further reduces efficiency degradation of the tracker module **1F**.

[0288] In one exemplary configuration of the tracker module **1E**, a switch included in the switched-capacitor circuit **20** and a switch included in the pre-regulator circuit **10** are connected to each other via the conductor wiring line **96** provided at the module laminate **90**.

[0289] According to the above-mentioned configuration, the wiring for connecting the PR switch portion **10A** and the SC switch portion **20A** can be provided at the module laminate **90**. This

configuration improves the freedom of layout of the wiring and achieve low impedance. Therefore, the signal transmission loss between the PR switch portion **10A** and the SC switch portion **20A** is reduced.

[0290] In one exemplary configuration of the tracker module **1E**, the at least one capacitor included in the switched-capacitor circuit **20** is disposed adjacent to the integrated circuit **80D** (or **80E**).

[0291] According to the above-mentioned configuration, the wiring for connecting the PR switch portion **10A** and the SC switch portion **20A** can be provided at the module laminate **90**. This configuration improves the freedom of layout of the wiring and achieve low impedance. Therefore, the signal transmission loss between the pre-regulator circuit **10** and the switched-capacitor circuit **20** is reduced.

[0292] The tracker modules **1A**, and **1C** to **1F** according to Example and its modifications include the module laminate **90**, and a single integrated circuit **80** (or **80C**, **80D**, **80E**, or **80F**) disposed at the module laminate **90**. The integrated circuit **80** (or **80C**, **80D**, **80E**, or **80F**) includes: a first switch, a second switch, a third switch, a fourth switch, a fifth switch, a sixth switch, a seventh switch, and an eighth switch that are included in the switched-capacitor circuit **20**; a ninth switch and a tenth switch that are included in the output switching circuit **30** connected to the switched-capacitor circuit **20**; an eleventh switch and a twelfth switch that are included in the pre-regulator circuit **10** connected to the switched-capacitor circuit **20**; and the digital control circuit **60** configured to control the first to twelfth switches. The switched-capacitor circuit **20** includes: a first capacitor having a first electrode and a second electrode; a second capacitor having a third electrode and a fourth electrode; and the first to eighth switches. One terminal of the first switch and one terminal of the third switch are connected to the first electrode. One terminal of the second switch and one terminal of the fourth switch are connected to the second electrode. One terminal of the fifth switch and one terminal of the seventh switch are connected to the third electrode. One terminal of the sixth switch and one terminal of the eighth switch are connected to the fourth electrode. Another terminal (e.g., a second terminal) of the first switch, another terminal of the second switch, another terminal of the fifth switch, and another terminal of the sixth switch are connected to each other. Another terminal of the third switch is connected to an other terminal of the seventh switch. Another terminal of the fourth switch is connected to another terminal of the eighth switch. The output switching circuit **30** includes: a first output terminal; the ninth switch connected between the first output terminal, and the other terminal of the first switch, the other terminal of the second switch, the other terminal of the fifth switch, and the other terminal of the sixth switch; and the tenth switch connected between the first output terminal, and the other terminal of the third switch and the other terminal of the seventh switch. The pre-regulator circuit **10** includes: an input terminal; the eleventh switch connected between the input terminal and one end of a power inductor; and the twelfth switch connected between the one end of the power inductor and ground. Another end of the power inductor is connected to the other terminal of the first switch, the other terminal of the second switch, the other terminal of the fifth switch, and the other terminal of the sixth switch. In plan view of the module laminate **90**, at least one of: the ninth switch and the tenth switch; or the digital control circuit **60** is disposed between: the eleventh switch and the twelfth switch; and the first switch, the second switch, the third switch, the fourth switch, the fifth switch, the sixth switch, the seventh switch, and the eighth switch.

[0293] According to the above-mentioned configuration, at least one of the digital control circuit **60** or the OS switch portion **30A**, which generates a relatively small amount of heat, is disposed between the PR switch portion **10A** and the SC switch portion **20A**, which are sources of heat generation. This configuration distributes and dissipates the heat generated by the PR switch portion **10A** and the SC switch portion **20A**. Therefore, efficient heat dissipation with reduced localized temperature rise is achieved. This configuration reduces efficiency degradation of the tracker modules **1A**, and **1C** to **1F**.

[0294] In one exemplary configuration of the tracker modules **1E** and **1F**, the first to eighth

switches are disposed adjacent to the ninth and tenth switches.

[0295] The above-mentioned configuration connects the SC switch portion **20A** and the OS switch portion **30A** by low-impedance wiring. This configuration improves transient response performance at the rising and falling voltage edges. Further, the low impedance of the above-mentioned wiring serving as a power line also reduce power consumption.

[0296] In one exemplary configuration of the tracker module **1C**, in plan view of the module laminate **90**, the first to twelfth switches are disposed in a peripheral region within the integrated circuit **80C**.

[0297] The above-mentioned configuration shortens the heat dissipation path from the PR switch portion **10A** and the SC switch portion **20A** to an external circuit connected to the tracker module **1C**, and consequently improves heat dissipation from the tracker module **1C**. The above-mentioned configuration also shortens the connection wiring between the PR switch portion **10A** and the capacitors included in the pre-regulator circuit **10**, and shortens the connection wiring between the SC switch portion **20A** and the capacitors included in the switched-capacitor circuit **20**. This configuration reduces signal transmission loss.

[0298] In one exemplary configuration of the tracker module **1C**, in plan view of the module laminate **90**, the digital control circuit **60** is disposed more centrally within the integrated circuit **80C** than are the first to twelfth switches.

[0299] According to the above-mentioned configuration, a sufficient distance can be provided between the PR switch portion **10A** and the SC switch portion **20A**, which are sources of heat generation. This configuration distributes and dissipates the heat generated by the PR switch portion **10A** and the SC switch portion **20A**. The above-mentioned configuration also shortens the control signal wiring that connects the digital control portion **60A** with each of the PR switch portion **10A**, the SC switch portion **20A**, and the OS switch portion **30A**. This configuration enables the digital control circuit **60** to accurately control the switching action of each switch portion.

[0300] In one exemplary configuration of the tracker modules **1D** and **1E**, in plan view of the module laminate **90**, the first to eighth switches are provided in a region formed by two elongated regions extending in mutually intersecting directions, and the two elongated regions are positioned along two peripheral sides of the integrated circuit **80D** (or **80E**).

[0301] The above-mentioned configuration shortens the connection wiring between the SC switch portion **20A** and the capacitors included in the switched-capacitor circuit **20**. This configuration also reduces heat generation in the switched-capacitor circuit **20**, and also reduces signal transmission loss.

[0302] In one exemplary configuration of the tracker module **1E**, at least one of the first to eighth switches, and at least one of the eleventh or twelfth switches are connected to each other via the conductor wiring line **96** provided at the module laminate **90**.

[0303] According to the above-mentioned configuration, the conductor wiring line for connecting the PR switch portion **10A** and the SC switch portion **20A** can be provided at the module laminate **90**. This configuration improves the freedom of layout of the conductor wiring line and achieve low impedance. Therefore, the signal transmission loss between the pre-regulator circuit **10** and the switched-capacitor circuit **20** is reduced.

Additional Exemplary Embodiments

[0304] Although the tracker module according to the exemplary aspects of the present disclosure has been described above based on embodiments, the embodiments are not intended to limit the tracker module. The exemplary aspects of the present disclosure are intended to also encompass: other embodiments implemented by combining any constituent elements in the above embodiments; modifications obtained by modifying the above embodiments in various ways as may become apparent to those skilled in the art without departing from the scope of the exemplary aspects; and various devices incorporating the tracker module mentioned above.

[0305] For example, in the circuit configurations of the various circuits according to the above embodiments, another circuit element, a wiring line, and other features may be inserted between individual circuit elements and paths connecting signal paths disclosed in the drawings. For example, an impedance matching circuit may be inserted between the power amplifier 2 and the filter 3.

[0306] According to the above exemplary embodiments, four discrete voltages are supplied to the power amplifier. The number of discrete voltages, however, is not limited to four. For example, improved power-added efficiency can be achieved as long as a plurality of discrete voltages include at least the following voltages: a voltage corresponding to the maximum output power; and a voltage corresponding to the output power with the highest frequency of occurrence.

[0307] The exemplary aspects of the present disclosure can be used for a wide variety of communication devices, such as mobile phones, as a tracker module that supplies voltage to a power amplifier.

REFERENCE SIGNS LIST

[0308] **1** tracker circuit [0309] **1A, 1B, 1C, 1D, 1E, 1F, 1G** tracker module [0310] **2** power amplifier [0311] **3** filter [0312] **4** antenna [0313] **5** RFIC [0314] **6** communication device [0315] **10** pre-regulator circuit [0316] **10A** PR switch portion [0317] **20** switched-capacitor circuit [0318] **20A** SC switch portion [0319] **30** output switching circuit [0320] **30A** OS switch portion [0321] **40** filter circuit [0322] **40A** FL switch portion [0323] **50** DC power source [0324] **60** digital control circuit [0325] **60A** digital control portion [0326] **61** first controller [0327] **62** second controller [0328] **80, 80C, 80D, 80E, 80F, 81B, 81G, 82B, 82G, 83B, 83G, 84B, 84G, 86B, 86G** integrated circuit [0329] **90** module laminate [0330] **90a, 90b** major face [0331] **91, 92** resin member [0332] **96, 97** conductor wiring line [0333] **110, 131, 132, 133, 134, 140** input terminal [0334] **111, 112, 113, 114, 130, 141** output terminal [0335] **150** external connection terminal [0336] **601, 602, 603, 604** control terminal

Claims

1. A tracker module comprising: a module laminate; and an integrated circuit disposed at the module laminate, the integrated circuit including: at least one switch included in a converter circuit that is configured to convert an input voltage into a first voltage, at least one switch included in a switched-capacitor circuit that is configured to generate a plurality of discrete second voltages based on the first voltage, at least one switch included in an output switching circuit that is configured to selectively output at least one voltage of the plurality of discrete second voltages generated by the switched-capacitor circuit, and a digital control circuit configured to, based on a digital control signal corresponding to an envelope signal, control the output switching circuit to selectively output the at least one voltage of the plurality of discrete second voltages, wherein in a plan view of the module laminate, at least one of: (i) the at least one switch included in the output switching circuit and (ii) the digital control circuit, is disposed between the at least one switch included in the converter circuit and the at least one switch included in the switched-capacitor circuit.

2. The tracker module according to claim 1, wherein the at least one switch included in the switched-capacitor circuit is disposed adjacent to the at least one switch included in the output switching circuit.

3. The tracker module according to claim 1, wherein, in the plan view of the module laminate, the at least one switch included in the output switching circuit, and the digital control circuit are disposed between the at least one switch included in the converter circuit and the at least one switch included in the switched-capacitor circuit.

4. The tracker module according to claim 1, wherein, in the plan view of the module laminate, the at least one switch included in the converter circuit, the at least one switch included in the

switched-capacitor circuit, and the at least one switch included in the output switching circuit are disposed in a peripheral region within the integrated circuit.

5. The tracker module according to claim 4, wherein, in the plan view of the module laminate, the digital control circuit is disposed more centrally within the integrated circuit than the at least one switch included in the converter circuit, the at least one switch included in the switched-capacitor circuit, and the at least one switch included in the output switching circuit.

6. The tracker module according to claim 1, wherein, in the plan view of the module laminate, the at least one switch included in the switched-capacitor circuit is in a region defined by two elongated regions that extend in mutually intersecting directions, and the two elongated regions are disposed along two peripheral sides of the integrated circuit.

7. The tracker module according to claim 6, wherein the at least one capacitor included in the switched-capacitor circuit is disposed adjacent to the integrated circuit.

8. The tracker module according to claim 4, wherein the at least one switch included in the switched-capacitor circuit is disposed separately in a first region and a second region that are spaced apart from each other.

9. The tracker module according to claim 8, wherein, in the plan view of the module laminate, the digital control circuit is disposed between the first region and the second region.

10. The tracker module according to claim 1, wherein a switch included in the switched-capacitor circuit is connected to a switch included in the converter circuit via an electrically conductive member at the module laminate.

11. The tracker module according to claim 10, wherein the at least one capacitor included in the switched-capacitor circuit is disposed adjacent to the integrated circuit.

12. A tracker module comprising: a module laminate; a first integrated circuit disposed at the module laminate and that includes at least one switch in a converter circuit that is configured to convert an input voltage into a first voltage; a second integrated circuit disposed at the module laminate and that includes at least one switch included in a switched-capacitor circuit that is configured to generate a plurality of discrete second voltages based on the first voltage; a third integrated circuit disposed at the module laminate and that includes at least one switch included in an output switching circuit that is configured to selectively output at least one voltage of the plurality of discrete second voltages generated by the switched-capacitor circuit; and a digital control circuit configured to, based on a digital control signal corresponding to an envelope signal, control the output switching circuit to selectively output at least one voltage of the plurality of discrete second voltages, wherein, in a plan view of the module laminate, at least one of: (i) the at least one switch included in the output switching circuit, and (ii) the digital control circuit, is disposed between the at least one switch included in the converter circuit and the at least one switch included in the switched-capacitor circuit.

13. The tracker module according to claim 12, wherein: the module laminate has a first major face and a second major face that are opposite to each other, an external connection terminal of the tracker module further is disposed at the second major face, and the first integrated circuit and the second integrated circuit are disposed at the second major face.

14. The tracker module according to claim 13, wherein: the third integrated circuit is disposed at the first major face, and the second integrated circuit at least partially overlaps the third integrated circuit in the plan view of the module laminate.

15. A tracker module comprising: a module laminate; and a integrated circuit disposed at the module laminate and that includes: a switched-capacitor circuit that includes a first switch, a second switch, a third switch, a fourth switch, a fifth switch, a sixth switch, a seventh switch, and an eighth switch, an output switching circuit that includes a ninth switch and a tenth switch that are connected to the switched-capacitor circuit, a converter circuit that includes an eleventh switch and a twelfth switch that are connected to the switched-capacitor circuit, and a digital control circuit configured to control the first to twelfth switches, wherein the switched-capacitor circuit includes: a

first capacitor having a first electrode and a second electrode, a second capacitor having a third electrode and a fourth electrode, and the first to eighth switches, wherein a first terminal of the first switch and a first terminal of the third switch are connected to the first electrode, wherein a first terminal of the second switch and a first terminal of the fourth switch are connected to the second electrode, wherein a first terminal of the fifth switch and a first terminal of the seventh switch are connected to the third electrode, wherein a first terminal of the sixth switch and a first terminal of the eighth switch are connected to the fourth electrode, wherein a second terminal of the first switch, a second terminal of the second switch, a second terminal of the fifth switch, and a second terminal of the sixth switch are connected to each other, wherein a second terminal of the third switch is connected to a second terminal of the seventh switch, wherein a second terminal of the fourth switch is connected to a second terminal of the eighth switch, wherein the output switching circuit includes: a first output terminal, the ninth switch connected between the first output terminal, and the second terminal of the first switch, the second terminal of the second switch, the second terminal of the fifth switch, and the second terminal of the sixth switch, and the tenth switch connected between the first output terminal, and the second terminal of the third switch and the second terminal of the seventh switch, wherein the converter circuit includes: an input terminal, the eleventh switch connected between the input terminal and a first end of a power inductor, and the twelfth switch connected between the first end of the power inductor and ground, wherein a second end of the power inductor is connected to the second terminal of the first switch, the second terminal of the second switch, the second terminal of the fifth switch, and the second terminal of the sixth switch, and wherein in a plan view of the module laminate, at least one of: (i) the ninth switch and the tenth switch, and (ii) the digital control circuit, is disposed between: the eleventh switch and the twelfth switch; and the first switch, the second switch, the third switch, the fourth switch, the fifth switch, the sixth switch, the seventh switch, and the eighth switch.

16. The tracker module according to claim 15, wherein the first switch, the second switch, the third switch, the fourth switch, the fifth switch, the sixth switch, the seventh switch, and the eighth switch are each disposed adjacent to the ninth switch and the tenth switch.

17. The tracker module according to claim 15, wherein, in the plan view of the module laminate, the first switch, the second switch, the third switch, the fourth switch, the fifth switch, the sixth switch, the seventh switch, the eighth switch, the ninth switch, the tenth switch, the eleventh switch, and the twelfth switch are disposed in a peripheral region within the integrated circuit.

18. The tracker module according to claim 17, wherein, in the plan view of the module laminate, the digital control circuit is disposed more centrally within the integrated circuit than the first switch, the second switch, the third switch, the fourth switch, the fifth switch, the sixth switch, the seventh switch, the eighth switch, the ninth switch, the tenth switch, the eleventh switch, and the twelfth switch.

19. The tracker module according to claim 15, wherein, in the plan view of the module laminate, the first switch, the second switch, the third switch, the fourth switch, the fifth switch, the sixth switch, the seventh switch, and the eighth switch are disposed in a region defined by two elongated regions extending in mutually intersecting directions, and the two elongated regions are disposed along two peripheral sides of the integrated circuit.

20. The tracker module according to claim 15, wherein at least one of the first switch, the second switch, the third switch, the fourth switch, the fifth switch, the sixth switch, the seventh switch, and the eighth switch, and at least one of the eleventh switch and the twelfth switch are connected to each other via an electrically conductive member provided at the module laminate.
